

Multi-Certificate Safety and Contingency Planning for Mars-Analog Helicopter Landing

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I. INTRODUCTION

Aerial vehicles must navigate and land with limited opportunities for external intervention, incomplete terrain knowledge, and no dependence on terrestrial positioning infrastructure. In this setting, autonomous landing is not only a trajectory-generation problem: the vehicle must retain reliable state estimation, avoid irregular obstacles, converge to a feasible touchdown state, and preserve alternative landing sites when the preferred site becomes unavailable. Terrain-dependent visual observability further couples motion planning to estimation performance, particularly when visual-inertial odometry is used over weakly textured terrain [1], [2].

Terrain-Aware Model Predictive Path Integral (TA-MPPI) planning addresses this coupling by incorporating expected visual-feature quality into the finite-horizon cost, thereby producing a perception-aware nominal command that balances progress toward a mission objective with informative camera viewpoints [2]. Such a planner improves nominal behavior, but a soft trajectory cost does not by itself establish forward invariance of a collision-free set. Moreover, committing to a

single target can drive the vehicle outside the states from which backup landing sites remain dynamically recoverable.

Control Barrier Functions (CBFs) provide a complementary mechanism for safety by restricting the control input so that a prescribed zero-superlevel set is forward invariant [3], [4]. Their practical use in complex environments, however, depends on constructing a sufficiently smooth safety function from perception-derived geometry. Poisson safety functions address this synthesis problem by solving a Dirichlet boundary-value problem over the free-space domain, producing a differentiable field whose zero level set represents obstacle boundaries and whose derivatives can be used in CBF or High-Order CBF constraints [5], [6]. Recent extensions have incorporated configuration-dependent geometry, prediction, and layered filtering, illustrating the compatibility of Poisson certificates with finite-horizon and real-time control architectures [7], [8].

Collision avoidance alone does not guarantee successful landing. Control Lyapunov Functions (CLFs) certify convergence toward controlled equilibria, and suitable CLF sublevel sets provide inner approximations of their local regions of attraction. Steering with contingencies uses these sets to formalize an r -out-of- p requirement: the system approaches an active equilibrium while remaining inside the certified attraction regions of at least r among p candidates [9]. Combinatorial CBF constructions enforce this logical requirement with a number of smooth constraints that scales linearly with the number of candidates, rather than with the $\binom{p}{r}$ possible combinations [10].

This work develops a unified multi-certificate formulation for Mars-analog helicopter landing. TA-MPPI supplies the perception-aware nominal input; an occupancy-derived Poisson function supplies the environmental CBF/HOCBF; a local CLF is associated with each candidate landing equilibrium; and the corresponding CLF sublevel certificates define an r -out-of- p combinatorial region of attraction. A single optimization-based filter then computes the closest admissible command while enforcing collision avoidance, active-target convergence, input constraints, and preservation of landing alternatives.

The current scope is an offline, reduced-order Mars-analog study. The formal claims therefore concern the stated reduced-order dynamics and verified certificates; they do not yet establish full-order safety for the PX4 flight stack, hardware flight, or Martian aerodynamic operation.

II. BACKGROUND

Notation: Let $[p] := \{1, \dots, p\}$ and let C^k denote the set of k -times continuously differentiable functions. A continuous function $\alpha : [0, a) \rightarrow [0, \infty)$ is class- $\mathcal{K}$ if it is strictly increasing and $\alpha(0) = 0$. A continuous function $\gamma : (-b, a) \rightarrow \mathbb{R}$, with $a, b > 0$, is extended class- $\mathcal{K}$ ($\gamma \in \mathcal{K}_e$) if it is strictly increasing and $\gamma(0) = 0$. For a C^1 function $h : \mathbb{R}^n \rightarrow \mathbb{R}$ and vector fields $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$, the Lie derivatives are

$$L_{\mathbf{f}}h(\mathbf{x}) := Dh(\mathbf{x}) \cdot \mathbf{f}(\mathbf{x}), \quad L_{\mathbf{g}}h(\mathbf{x}) := Dh(\mathbf{x}) \cdot \mathbf{g}(\mathbf{x}),$$

where Dh and D^2h denote the gradient row and Hessian, respectively. For $r > 0$, $B_r(\mathbf{x}^*) := \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x} - \mathbf{x}^*\| < r\}$ is the open ball centered at $\mathbf{x}^*$. Additionally, $\text{ReLU}(s) := \max\{0, s\}$ and $\max^{(r)}\{s_j\}_{j \in [p]}$ denotes the r -th largest value among $\{s_j\}_{j \in [p]}$.

A. Control Lyapunov functions

Consider the nonlinear control-affine system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u}, \quad \mathbf{x} \in \mathbb{R}^n, \quad \mathbf{u} \in \mathcal{U} \subseteq \mathbb{R}^m, \quad (1)$$

where $\mathbf{f}$ and $\mathbf{g}$ are locally Lipschitz. Let $\mathbf{x}^*$ be a controlled equilibrium of (1), i.e., there exists $\mathbf{u}^* \in \mathcal{U}$ such that

$$\mathbf{f}(\mathbf{x}^*) + \mathbf{g}(\mathbf{x}^*)\mathbf{u}^* = 0. \quad (2)$$

Control Lyapunov functions characterize feedback inputs that stabilize $\mathbf{x}^*$ [11], [12], [13].

Definition 1 (Local control Lyapunov function). A C^1 function $V : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ is a local control Lyapunov function (CLF) for (1) about $\mathbf{x}^*$ if $V(\mathbf{x}^*) = 0$ and there exist class- $\mathcal{K}$ functions α_1 , α_2 , and α_V , together with $r_V > 0$, such that

$$\alpha_1(\|\mathbf{x} - \mathbf{x}^*\|) \leq V(\mathbf{x}) \leq \alpha_2(\|\mathbf{x} - \mathbf{x}^*\|), \quad (3a)$$

$$\inf_{\mathbf{u} \in \mathcal{U}} \dot{V}(\mathbf{x}, \mathbf{u}) \leq -\alpha_V(V(\mathbf{x})). \quad (3b)$$

Condition (3a) holds for every $\mathbf{x} \in B_{r_V}(\mathbf{x}^*)$, and (3b) holds for every $\mathbf{x} \in B_{r_V}(\mathbf{x}^*) \setminus \{\mathbf{x}^*\}$, where

$$\dot{V}(\mathbf{x}, \mathbf{u}) \triangleq L_{\mathbf{f}}V(\mathbf{x}) + L_{\mathbf{g}}V(\mathbf{x})\mathbf{u}. \quad (4)$$

◇

Conditions (3) imply that for every $\mathbf{x} \in B_{r_V}(\mathbf{x}^*) \setminus \{\mathbf{x}^*\}$ there exists an admissible input for which $\dot{V}(\mathbf{x}, \mathbf{u}) < 0$. The corresponding set of stabilizing controls is

$$\mathcal{K}_{\text{CLF}}(\mathbf{x}) := \left\{ \mathbf{u} \in \mathcal{U} \mid \dot{V}(\mathbf{x}, \mathbf{u}) \leq -\alpha_V(V(\mathbf{x})) \right\}. \quad (5)$$

Lemma 1 (CLF stabilization). Let V be a local CLF for (1), and let $\mathbf{k} : \mathbb{R}^n \rightarrow \mathcal{U}$ be a continuous feedback controller such that

$$\dot{V}(\mathbf{x}, \mathbf{k}(\mathbf{x})) \leq -\alpha_V(V(\mathbf{x})), \quad (6)$$

and $\mathbf{k}(\mathbf{x}^*) = \mathbf{u}^*$. Then $\mathbf{k}$ asymptotically stabilizes the equilibrium $\mathbf{x}^*$ on the domain where the CLF conditions hold. ■

Lyapunov functions also provide certified inner approximations of regions of attraction. For $c > 0$, define the sublevel set

$$\Omega_c := \{\mathbf{x} \in \mathbb{R}^n \mid V(\mathbf{x}) \leq c\}. \quad (7)$$

Lemma 2 (CLF sublevel set). Under the conditions of Lemma 1, let $c > 0$ be such that Ω_c is compact and $\Omega_c \subset B_{r_V}(\mathbf{x}^*)$. Then the feedback $\mathbf{u} = \mathbf{k}(\mathbf{x})$ renders Ω_c forward invariant. Moreover, Ω_c is a certified inner approximation of the region of attraction of $\mathbf{x}^*$. ■

Given a nominal controller $\mathbf{k}_{\text{nom}}$, the CLF condition can be enforced through the quadratic program

$$\begin{aligned} \mathbf{k}(\mathbf{x}) = \arg \min_{\mathbf{u} \in \mathcal{U}} \quad & \|\mathbf{u} - \mathbf{k}_{\text{nom}}(\mathbf{x})\|_2^2 \\ \text{s.t.} \quad & \dot{V}(\mathbf{x}, \mathbf{u}) \leq -\alpha_V(V(\mathbf{x})). \end{aligned} \quad (8)$$

This formulation preserves the nominal input whenever it already satisfies the CLF decrease condition and otherwise computes the nearest stabilizing input [4].

For each candidate landing equilibrium $(\mathbf{x}_j^*, \mathbf{u}_j^*)$, a quadratic local CLF may be constructed from the linearization

$$\delta \dot{\mathbf{x}}_j = A_j \delta \mathbf{x}_j + B_j \delta \mathbf{u}_j, \quad \delta \mathbf{x}_j = \mathbf{x} - \mathbf{x}_j^*, \quad \delta \mathbf{u}_j = \mathbf{u} - \mathbf{u}_j^*. \quad (9)$$

Choose $\delta \mathbf{u}_j = -K_j \delta \mathbf{x}_j$ such that $A_{\text{cl},j} = A_j - B_j K_j$ is Hurwitz. Given $Q_j = Q_j^\top \succ 0$, the Lyapunov equation

$$A_{\text{cl},j}^\top P_j + P_j A_{\text{cl},j} = -Q_j \quad (10)$$

has a unique solution $P_j = P_j^\top \succ 0$, and

$$V_j(\mathbf{x}) := (\mathbf{x} - \mathbf{x}_j^*)^\top P_j (\mathbf{x} - \mathbf{x}_j^*) \quad (11)$$

is a Lyapunov function for the linear closed-loop model. By smoothness of the nonlinear dynamics, V_j remains a local CLF on a sufficiently small neighborhood of $\mathbf{x}_j^*$ [9], [11].

B. Control barrier functions and high-order constraints

While CLFs enforce convergence to an equilibrium, Control Barrier Functions enforce forward invariance of a set. Let

$$\mathcal{C} := \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) \geq 0\}, \quad (12a)$$

$$\partial \mathcal{C} := \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) = 0\}, \quad (12b)$$

$$\text{int}(\mathcal{C}) := \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) > 0\}, \quad (12c)$$

where $h : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable. A set is forward invariant if every trajectory initialized in the set remains in it for all future time. For a regular boundary, Nagumo's theorem requires

$$\dot{h}(\mathbf{x}) = Dh(\mathbf{x}) \cdot (\mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{k}(\mathbf{x})) \geq 0, \quad \forall \mathbf{x} \in \partial \mathcal{C}. \quad (13)$$

Definition 2 (Control Barrier Function). Let $\mathcal{C}$ be the zero-superlevel set of a C^1 function $h : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying $Dh(\mathbf{x}) \neq 0$ when $h(\mathbf{x}) = 0$. The function h is a control barrier function (CBF) for system (1) if there exists $\gamma \in \mathcal{K}_e$ such that

$$\sup_{\mathbf{u} \in \mathcal{U}} \left\{ \dot{h}(\mathbf{x}, \mathbf{u}) = L_{\mathbf{f}}h(\mathbf{x}) + L_{\mathbf{g}}h(\mathbf{x})\mathbf{u} > -\gamma(h(\mathbf{x})) \right\} \quad (14)$$

for all $\mathbf{x}$ in a domain containing $\mathcal{C}$. ◇

The set of pointwise safe controls is

$$\mathcal{K}_{\text{CBF}}(\mathbf{x}) := \left\{ \mathbf{u} \in \mathcal{U} \mid \dot{h}(\mathbf{x}, \mathbf{u}) \geq -\gamma(h(\mathbf{x})) \right\}. \quad (15)$$

Lemma 3 (CBF forward invariance). Let h define $\mathcal{C}$ as in (12), and let $\mathbf{k} : \mathbb{R}^n \rightarrow \mathcal{U}$ be a locally Lipschitz controller satisfying

$$\dot{h}(\mathbf{x}, \mathbf{k}(\mathbf{x})) \geq -\gamma(h(\mathbf{x})) \quad (16)$$

on a neighborhood of $\mathcal{C}$. Then $\mathbf{u} = \mathbf{k}(\mathbf{x})$ renders $\mathcal{C}$ forward invariant. ■

Given a desired, potentially unsafe controller $\mathbf{k}_{\text{nom}}$, the CBF safety filter is

$$\begin{aligned} \mathbf{k}(\mathbf{x}) &= \arg \min_{\mathbf{u} \in \mathcal{U}} \|\mathbf{u} - \mathbf{k}_{\text{nom}}(\mathbf{x})\|_2^2 \\ \text{s.t. } L_{\mathbf{f}}h(\mathbf{x}) + L_{\mathbf{g}}h(\mathbf{x})\mathbf{u} &\geq -\gamma(h(\mathbf{x})). \end{aligned} \quad (\text{Safety-Filter})$$

Thus, the nominal action is modified only when it lies outside $\mathcal{K}_{\text{CBF}}(\mathbf{x})$ [3], [4].

Relative degree and integrator chains. Let $\mathbf{y} : \mathbb{R}^n \rightarrow \mathbb{R}^q$ be a vector output. The output has relative degree $r \in \mathbb{N}$ for (1) if

$$L_{\mathbf{g}}L_{\mathbf{f}}^i\mathbf{y}(\mathbf{x}) = 0, \quad \forall i \in \{0, \dots, r-2\}, \quad (18a)$$

$$\text{rank}(L_{\mathbf{g}}L_{\mathbf{f}}^{r-1}\mathbf{y}(\mathbf{x})) = q, \quad \forall \mathbf{x} \in \mathbb{R}^n. \quad (18b)$$

Define the partial coordinates

$$\vec{\mathbf{y}}(\mathbf{x}) := \begin{bmatrix} \mathbf{y}(\mathbf{x}) \\ \mathbf{y}^{(1)}(\mathbf{x}) \\ \vdots \\ \mathbf{y}^{(r-1)}(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \mathbf{y}(\mathbf{x}) \\ L_{\mathbf{f}}\mathbf{y}(\mathbf{x}) \\ \vdots \\ L_{\mathbf{f}}^{r-1}\mathbf{y}(\mathbf{x}) \end{bmatrix}. \quad (19)$$

These coordinates satisfy the chain-of-integrators dynamics

$$\frac{d}{dt}\vec{\mathbf{y}} = \underbrace{\begin{bmatrix} 0 & I_{q(r-1)} \\ 0 & 0 \end{bmatrix}}_{A_r} \vec{\mathbf{y}} + \underbrace{\begin{bmatrix} 0 \\ I_q \end{bmatrix}}_{B_r} \mathbf{w}, \quad (20)$$

where

$$\mathbf{w} := L_{\mathbf{f}}^r\mathbf{y}(\mathbf{x}) + L_{\mathbf{g}}L_{\mathbf{f}}^{r-1}\mathbf{y}(\mathbf{x})\mathbf{u}. \quad (21)$$

This representation identifies the differentiation layer at which the control affects a safety output [5], [14].

High-Order Control Barrier Functions. Let $h_0 := h$ have relative degree $r \geq 2$. Define recursively

$$h_i(\mathbf{x}) := \dot{h}_{i-1}(\mathbf{x}) + \alpha_i(h_{i-1}(\mathbf{x})), \quad i = 1, \dots, r-1, \quad (22)$$

where $\alpha_i \in \mathcal{K}_e$ have sufficient differentiability, and define

$$\mathcal{C}_i := \{\mathbf{x} \in \mathbb{R}^n \mid h_i(\mathbf{x}) \geq 0\}. \quad (23)$$

The function $h_H := h_{r-1}$ is a HOCBF if there exists $\alpha_r \in \mathcal{K}_e$ such that

$$\dot{h}_H(\mathbf{x}, \mathbf{u}) \geq -\alpha_r(h_H(\mathbf{x})). \quad (24)$$

Lemma 4 (Higher-order forward invariance). If h_H is a HOCBF and a locally Lipschitz controller satisfies (24), then

$$\mathcal{C}_H := \bigcap_{i=0}^{r-1} \mathcal{C}_i \quad (25)$$

is forward invariant. ■

For the double-integrator model

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{a}, \quad (26)$$

and a spatial safety function $h_P(\mathbf{p})$,

$$\dot{h}_P(\mathbf{p}, \mathbf{v}) = Dh_P(\mathbf{p})\mathbf{v}, \quad (27a)$$

$$\ddot{h}_P(\mathbf{p}, \mathbf{v}, \mathbf{a}) = Dh_P(\mathbf{p})\mathbf{a} + \mathbf{v}^\top D^2h_P(\mathbf{p})\mathbf{v}. \quad (27b)$$

With $\alpha_i(s) = \gamma_i s$, $\gamma_i > 0$, the relative-degree-two HOCBF constraint becomes

$$\begin{aligned} Dh_P(\mathbf{p})\mathbf{a} &\geq -\mathbf{v}^\top D^2h_P(\mathbf{p})\mathbf{v} \\ &\quad - (\gamma_1 + \gamma_2)Dh_P(\mathbf{p})\mathbf{v} - \gamma_1\gamma_2h_P(\mathbf{p}). \end{aligned} \quad (28)$$

The inequality is affine in the acceleration input and can therefore be included directly in a QP [6].

Safety and stabilization may be combined by relaxing only the CLF row:

$$\begin{aligned} (\mathbf{u}^*, \delta^*) &= \arg \min_{\mathbf{u}, \delta} \frac{1}{2}\|\mathbf{u} - \mathbf{u}_{\text{nom}}\|_R^2 + p_\delta \delta^2 \\ \text{s.t. } \dot{V}(\mathbf{x}, \mathbf{u}) &\leq -\alpha_V(V(\mathbf{x})) + \delta, \\ \dot{h}(\mathbf{x}, \mathbf{u}) &\geq -\gamma(h(\mathbf{x})), \\ \mathbf{u} &\in \mathcal{U}. \end{aligned} \quad (29)$$

The hard CBF preserves safety, while δ permits temporary loss of Lyapunov decrease when safety and convergence are incompatible [3].

C. Combinatorial regions of attraction

Let $\{\mathbf{x}_j^*\}_{j \in [p]}$ be candidate landing equilibria, with local CLFs $\{V_j\}_{j \in [p]}$. For each j , choose $c_j > 0$ such that

$$\mathcal{R}_j := \{\mathbf{x} \in \mathbb{R}^n \mid V_j(\mathbf{x}) \leq c_j\} \quad (30)$$

is a certified inner approximation of the region of attraction of $\mathbf{x}_j^*$. Define

$$h_j(\mathbf{x}) := c_j - V_j(\mathbf{x}), \quad \mathcal{R}_j = \{\mathbf{x} \mid h_j(\mathbf{x}) \geq 0\}. \quad (31)$$

The r -out-of- p combinatorial region of attraction is

$$\tilde{\mathcal{R}}_r := \{\mathbf{x} \in \mathbb{R}^n \mid |\{j \in [p] \mid \mathbf{x} \in \mathcal{R}_j\}| \geq r\}. \quad (32)$$

Equivalently, define the pivot function

$$\tilde{h}_r(\mathbf{x}) := \max_{j \in [p]}^{(r)} \{h_j(\mathbf{x})\}, \quad (33)$$

so that

$$\tilde{\mathcal{R}}_r = \{\mathbf{x} \in \mathbb{R}^n \mid \tilde{h}_r(\mathbf{x}) \geq 0\}. \quad (34)$$

The cases $r = p$ and $r = 1$ recover logical conjunction and disjunction, respectively. Because the order statistic in (33) is nonsmooth where certificates exchange rank, it is not differentiated directly.

Lemma 5 (Combinatorial forward invariance). Suppose there exist a continuous controller $\mathbf{k} : \mathbb{R}^n \rightarrow \mathcal{U}$ and a nonnegative auxiliary function $\theta : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\begin{aligned} \dot{h}_j(\mathbf{x}, \mathbf{k}(\mathbf{x})) &\geq -\alpha_c(h_j(\mathbf{x})) \\ &\quad - \theta(\mathbf{x})\rho(h_j(\mathbf{x}) - \tilde{h}_r(\mathbf{x})), \end{aligned} \quad (35)$$

for all $j \in [p]$, where $\alpha_c \in \mathcal{K}_e$ and $\rho : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ is positive definite. Then $\mathbf{k}$ renders $\tilde{\mathcal{R}}_r$ forward invariant. ■

The controller and auxiliary function can be computed from

$$\begin{aligned} (\mathbf{k}(\mathbf{x}), \theta(\mathbf{x})) = \arg \min_{\mathbf{u}, \omega} & \frac{1}{2} \|\mathbf{u} - \mathbf{k}_{\text{nom}}(\mathbf{x})\|_2^2 + c_\omega \omega^2 \\ \text{s.t.} & \quad \dot{h}_j(\mathbf{x}, \mathbf{u}) \geq -\alpha_c(h_j(\mathbf{x})) \\ & \quad - \omega \rho(h_j(\mathbf{x}) - \tilde{h}_r(\mathbf{x})), \\ & \quad \omega \geq 0, \quad \mathbf{u} \in \mathcal{U}. \end{aligned} \quad (36)$$

where the certificate inequality is imposed for every $j \in [p]$. Together with one CLF constraint for the selected equilibrium, the combinatorial stabilization filter uses $p+1$ certificate rows and avoids enumerating the $\binom{p}{r}$ possible target combinations [9], [10]. An additional Poisson CBF or HOCBF row may subsequently be included to enforce collision avoidance with the physical environment.

D. Poisson safety functions

Consider systems whose spatial safety specifications are expressed through the output

$$\mathbf{y} := \mathbf{y}(\mathbf{x}) = (x, y, z) \in \mathbb{R}^3. \quad (37)$$

Let $\Omega \subset \mathbb{R}^3$ be an open, bounded, and connected free-space domain with boundary $\partial\Omega$ corresponding to obstacle surfaces and, when included, the external workspace boundary.

Definition 3 (Spatial safety function). A function $h_P : \bar{\Omega} \rightarrow \mathbb{R}$ is a safety function of order k on Ω if h_P is k -times differentiable, $Dh_P(\mathbf{y}) \neq 0$ when $h_P(\mathbf{y}) = 0$, and

$$\mathcal{C}_P = \{\mathbf{y} \in \bar{\Omega} \mid h_P(\mathbf{y}) \geq 0\}, \quad (38a)$$

$$\partial\mathcal{C}_P = \{\mathbf{y} \in \bar{\Omega} \mid h_P(\mathbf{y}) = 0\}, \quad (38b)$$

$$\text{int}(\mathcal{C}_P) = \{\mathbf{y} \in \bar{\Omega} \mid h_P(\mathbf{y}) > 0\}. \quad (38c)$$

◇

A Poisson safety function is synthesized by solving the Dirichlet problem

$$\begin{cases} \Delta h_P(\mathbf{y}) = f_P(\mathbf{y}), & \mathbf{y} \in \Omega, \\ h_P(\mathbf{y}) = 0, & \mathbf{y} \in \partial\Omega, \end{cases} \quad (39)$$

where

$$\Delta h_P := \frac{\partial^2 h_P}{\partial x^2} + \frac{\partial^2 h_P}{\partial y^2} + \frac{\partial^2 h_P}{\partial z^2} \quad (40)$$

and $f_P : \Omega \rightarrow \mathbb{R}$ is the forcing function. The geometry of free space is encoded by Ω and its zero Dirichlet boundary, whereas f_P determines the interior curvature of the solution.

Theorem 1 (Poisson safety function). Let $\Omega \subset \mathbb{R}^3$ be open, bounded, and connected, with boundary regular enough for the classical Dirichlet problem. Suppose $f_P \in C^{k,\alpha}(\bar{\Omega}; \mathbb{R}_{<0})$ for some $k \in \mathbb{N}_0$ and $\alpha \in (0, 1)$. Then the unique solution of (39) satisfies

$$h_P \in C^{k+2,\alpha}(\bar{\Omega}; \mathbb{R}_{\geq 0}), \quad (41)$$

with

$$h_P(\mathbf{y}) > 0 \quad \forall \mathbf{y} \in \Omega, \quad h_P(\mathbf{y}) = 0 \quad \forall \mathbf{y} \in \partial\Omega, \quad (42)$$

and $Dh_P(\mathbf{y}) \neq 0$ at regular boundary points. Hence h_P is a safety function of order $k+2$ whose zero-superlevel set is $\bar{\Omega}$. ■

The result follows from classical elliptic regularity, the minimum principle, and Hopf's boundary-point lemma; the required statements are collected in Appendix B [5], [15], [16]. For the single-integrator output dynamics $\dot{\mathbf{y}} = \mathbf{u}$, h_P directly yields

$$Dh_P(\mathbf{y}) \mathbf{u} \geq -\gamma(h_P(\mathbf{y})). \quad (43)$$

For acceleration-controlled motion, h_P , Dh_P , and D^2h_P enter the HOCBF inequality (28). Thus, the Poisson construction supplies a single differentiable geometric certificate for arbitrary occupancy geometry, while the CBF/HOCBF construction incorporates the system dynamics.

E. Terrain-aware perceptual planning

Terrain-Aware Model Predictive Path Integral (TA-MPPI) planning augments a finite-horizon nominal planner with the expected visual-feature quality of the terrain observed by the vehicle camera [2], [17]. Let $\mathcal{F}(\mathbf{x}) \subset \mathbb{R}^2$ denote the ground-plane projection of the camera footprint at state $\mathbf{x}$, let $\mathcal{Q}(\mathbf{p}) \in [0, 1]$ denote the expected feature quality at a ground point $\mathbf{p} = (x, y)$, and let $w(\mathbf{p})$ be a spatial confidence weight. The viewpoint score is

$$S(\mathbf{x}) = \frac{\int_{\mathcal{F}(\mathbf{x})} \mathcal{Q}(\mathbf{p}) w(\mathbf{p}) dA}{\int_{\mathcal{F}(\mathbf{x})} w(\mathbf{p}) dA}. \quad (44)$$

The denominator normalizes the spatial weights, so S is the weighted mean feature quality within the projected footprint.

For the simplified hover kinematics used by Todd et al., the state and control are

$$\mathbf{x}_k = \begin{bmatrix} \mathbf{p}_k \\ \psi_k \end{bmatrix} \in \mathbb{R}^3, \quad \mathbf{u}_k = \begin{bmatrix} \mathbf{v}_k \\ \omega_k \end{bmatrix} \in \mathbb{R}^3, \quad (45)$$

where $\mathbf{p}_k \in \mathbb{R}^2$ is planar position, ψ_k is heading, $\mathbf{v}_k \in \mathbb{R}^2$ is linear velocity, and ω_k is angular velocity. The finite-horizon problem is

$$\begin{aligned} \min_{\mathbf{u}_0:N-1} & \sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k, t_s) + V_f(\mathbf{x}_N) \\ \text{s.t.} & \quad \mathbf{x}_0 = \mathbf{s}, \\ & \quad \mathbf{x}_{k+1} = F(\mathbf{x}_k, \mathbf{u}_k), \quad 0 \leq k < N, \end{aligned} \quad (46)$$

with stage cost

$$\begin{aligned} \ell(\mathbf{x}_k, \mathbf{u}_k, t_s) = & \ell_{\text{goal}} + \ell_{\text{action}} + \ell_{\psi} \\ & + \ell_{\text{perception}} + \ell_{\text{boundary}}, \end{aligned} \quad (47)$$

where

$$\ell_{\text{perception}} = c_{\text{perception}}(1 - S_k(\mathbf{x})). \quad (48)$$

The remaining terms penalize distance to the goal, control effort, heading misalignment, and departure from the mapped region. In receding-horizon operation, the first optimized command

$$\mathbf{u}_{\text{nom}} := \mathbf{u}_0^* \quad (49)$$

provides the nominal task and perception command to the downstream certificate-based filter. Accordingly, the later formulation combines TA-MPPI nominal performance, a Poisson CBF/HOCBF for collision avoidance, an active-target CLF for convergence, and the combinatorial CLF sublevel certificates for preservation of landing alternatives.

III. PROBLEM FORMULATION

This section specializes the certificate objects introduced in Section II to a reduced-order Mars-analog landing problem. The objective is not merely to avoid obstacles or to reach one fixed waypoint. The vehicle must approach a selected landing equilibrium, retain a prescribed number of alternative equilibria, and react consistently when a candidate site is declared unavailable.

A. Reduced-order aerial-vehicle model

The controlled state is

$$\mathbf{x} := \begin{bmatrix} \mathbf{p} \\ \mathbf{v} \end{bmatrix} \in \mathbb{R}^6, \quad \mathbf{u} := \mathbf{a} \in \mathbb{R}^3, \quad (50)$$

where $\mathbf{p} = (p_x, p_y, p_z)$ is position, $\mathbf{v}$ is translational velocity, and $\mathbf{a}$ is a commanded acceleration. The reduced-order dynamics are the three-dimensional double integrator

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{a}, \quad \mathbf{A} = \begin{bmatrix} 0 & \mathbf{I}_3 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \mathbf{I}_3 \end{bmatrix}. \quad (51)$$

This model represents the task-level translational dynamics used by the certificate filter. It is deliberately separated from any future full-order attitude, thrust, estimator, or PX4 tracking dynamics. The reduced-order command is constrained by

$$-\mathbf{a}_{\max} \preceq \mathbf{a} \preceq \mathbf{a}_{\max}, \quad (52a)$$

$$-\mathbf{v}_{\max} \preceq \mathbf{v}_k + \Delta t \mathbf{a}_k \preceq \mathbf{v}_{\max}, \quad (52b)$$

where (52b) is a zero-order-hold one-step velocity constraint. These sampled bounds complement, but do not replace, the continuous-time certificate conditions.

B. Environment and landing candidates

Let $\mathcal{W} \subset \mathbb{R}^3$ be a bounded workspace represented by a binary occupancy grid O . After conservative obstacle inflation, the free-space domain and its boundary are

$$\Omega(O) := \{\mathbf{y} \in \mathcal{W} : O(\mathbf{y}) = 0\}, \quad \partial\Omega(O) := \partial\Omega. \quad (53)$$

The set $\partial\Omega$ contains obstacle surfaces and, when enabled, the outer computational boundary. Environmental safety is represented by a Poisson safety function h_P synthesized on this domain as described in Section II-D.

The mission provides p candidate landing sites. Candidate $j \in [p]$ has position $\mathbf{p}_j^*$ and controlled equilibrium

$$\mathbf{x}_j^* := \begin{bmatrix} \mathbf{p}_j^* \\ 0 \end{bmatrix}, \quad \mathbf{u}_j^* = 0. \quad (54)$$

A binary signal $\sigma_j(t) \in \{0, 1\}$ indicates whether site j is currently available. The available index set is

$$\mathcal{I}(t) := \{j \in [p] : \sigma_j(t) = 1\}, \quad (55)$$

and $j^\dagger(t) \in \mathcal{I}(t)$ denotes the actively pursued landing site. Availability is a mission-level statement; it is distinct from the state-dependent question of whether the vehicle lies inside a candidate CLF sublevel set.

C. Safety, convergence, and contingency specifications

The environmental safe set is

$$\mathcal{C}_P := \{\mathbf{x} \in \mathbb{R}^6 \mid h_P(\mathbf{p}) \geq 0\}. \quad (56)$$

Because h_P has relative degree two with respect to (51), forward invariance is enforced through the HOCBF construction in (28).

For each landing equilibrium, let V_j be a quadratic CLF and choose $c_j > 0$. Define the candidate attraction-region certificate

$$h_j^{\text{ROA}}(\mathbf{x}) := c_j - V_j(\mathbf{x}), \quad \mathcal{R}_j = \{\mathbf{x} : h_j^{\text{ROA}}(\mathbf{x}) \geq 0\}. \quad (57)$$

For a required contingency count r , with $1 \leq r \leq |\mathcal{I}(t)|$, define the available-site pivot

$$\tilde{h}_r^{\mathcal{I}}(\mathbf{x}, t) := \max^{(r)} \{h_j^{\text{ROA}}(\mathbf{x})\}_{j \in \mathcal{I}(t)}. \quad (58)$$

The corresponding combinatorial region is

$$\tilde{\mathcal{R}}_r^{\mathcal{I}}(t) := \{\mathbf{x} : \tilde{h}_r^{\mathcal{I}}(\mathbf{x}, t) \geq 0\}, \quad (59)$$

which contains the states lying inside at least r available candidate sublevel sets. This is the CLF-based combinatorial stabilization branch of the contingency framework, rather than a finite-horizon HJ reach-avoid construction [9].

Problem 1 (Multi-certificate contingency landing). Given O , the current state $\mathbf{x}$, candidate equilibria $\{\mathbf{x}_j^*\}_{j \in [p]}$, availability set $\mathcal{I}(t)$, an active site $j^\dagger(t)$, and a nominal acceleration $\mathbf{a}_{\text{nom}}$, find a locally Lipschitz filtered command $\mathbf{a}_{\text{safe}}$ and a target update policy such that:

- 1) the HOCBF environmental set is forward invariant;
- 2) acceleration and sampled velocity bounds are satisfied;
- 3) the active CLF decreases whenever compatible with the hard constraints;
- 4) the combinatorial set $\tilde{\mathcal{R}}_r^{\mathcal{I}}(t)$ is preserved while at least r available certificates remain; and
- 5) the modification $\|\mathbf{a}_{\text{safe}} - \mathbf{a}_{\text{nom}}\|$ is minimized.

If $|\mathcal{I}(t)| < r$, or if no available certified target exists, the controller must report loss of the requested contingency property rather than silently selecting an uncertified target.

The following assumptions delimit the present claims: the occupancy used by one Poisson solve is static; the reduced-order state is available; the Poisson field has sufficient numerical regularity for the required derivatives; each selected c_j satisfies the stated reduced-model and input-feasibility checks; and the multi-certificate program is feasible at the evaluated state. Full-order tracking, estimator delay, aerodynamic uncertainty, and intersample robustness are outside the current theoretical scope.

IV. APPROACH

The proposed architecture separates nominal performance from formal certificate generation. The planner supplies a desired reduced-order command; the Poisson, HOCBF, CLF, and contingency modules independently construct mathematical objects; and a final optimization layer reconciles the resulting constraints. This separation follows the minimally invasive CLF–CBF paradigm [3], [4] and the modular, perception-to-control structure used in recent Poisson safety-filter architectures [5], [8].

A. Modular certificate contracts

Every safety box exports affine rows using one canonical convention,

$$A_\ell \mathbf{z} \geq b_\ell, \quad (60)$$

where $\mathbf{z}$ is an augmented decision vector. A box does not solve the complete control problem and does not need to know how other certificates were constructed. This permits the environmental, stabilization, and contingency layers to be enabled, disabled, or replaced independently. Table I summarizes the scientific contracts.

B. Step 1: nominal command and active-site management

The architecture accepts any planner whose first command can be expressed in the reduced-order input space. A perception-aware planner such as TA–MPPI can therefore provide $\mathbf{a}_{\text{nom}}$ without changing any downstream certificate [2]. In the present numerical study, the nominal layer is instantiated by a clearance-weighted A* path followed by PD tracking. Let $\mathbf{p}_{\text{ref}}$ and $\mathbf{v}_{\text{ref}}$ denote the look-ahead reference extracted from the current path. The nominal command is

$$\mathbf{a}_{\text{nom}} = \text{sat}_{\bar{a}_{\text{nom}}} (k_p(\mathbf{p}_{\text{ref}} - \mathbf{p}) + k_v(\mathbf{v}_{\text{ref}} - \mathbf{v})), \quad (61)$$

where the saturation is applied before the multi-certificate filter. The straight line to a landing site is retained as an unsafe reference for geometric diagnostics, but is not used as the baseline nominal trajectory.

The active target remains fixed while it is available. If it becomes unavailable, the target manager selects the available certified candidate with the largest attraction margin,

$$j^\dagger \leftarrow \arg \max_{\substack{j \in \mathcal{I}(t) \\ h_j^{\text{ROA}}(\mathbf{x}) \geq 0}} h_j^{\text{ROA}}(\mathbf{x}). \quad (62)$$

If the set in (62) is empty, the controller reports that no available certified target exists. Target ranking is a mission decision; the safety and contingency claims continue to be supplied by the certificate rows, not by the ranking score.

C. Step 2: occupancy-to-Poisson safety synthesis

The environmental pipeline begins with conservative occupancy inflation to account for the vehicle footprint and a perception margin. The inflated binary map induces the free domain Ω and Dirichlet boundary $\partial\Omega$. The Poisson box then solves

$$\begin{cases} \Delta h_P(\mathbf{y}) = f_P(\mathbf{y}), & \mathbf{y} \in \Omega, \\ h_P(\mathbf{y}) = 0, & \mathbf{y} \in \partial\Omega, \end{cases} \quad (63)$$

with $f_P < 0$. Under the assumptions reviewed in Section II–D, the solution is positive in free space and has its zero level set on the obstacle boundary [5]. The forcing field changes the interior curvature and gradient distribution but does not change the encoded obstacle geometry. The implementation exposes constant, distance-weighted, average-flux, and guidance forcing methods. This choice is not cosmetic: the HOCBF samples h_P , Dh_P , and D^2h_P , so two solutions with the same zero boundary can induce different safe-control rows.

a) *Candidate forcing families.*: Let $d(\mathbf{y}) = \text{dist}(\mathbf{y}, \partial\Omega)$ and $d_{\text{max}} = \|d\|_{L^\infty(\Omega)}$. The first three implemented families are

$$f_c(\mathbf{y}) = -c, \quad c > 0, \quad (64a)$$

$$f_d(\mathbf{y}) = -\left(\frac{d(\mathbf{y})}{d_{\text{max}}}\right)^\eta, \quad 0 < \eta < 1, \quad (64b)$$

$$f_{\text{af}}(\mathbf{y}) = \bar{b} \frac{|\partial\Omega|}{|\Omega|}, \quad \bar{b} < 0. \quad (64c)$$

The average-flux expression follows from the divergence theorem,

$$\int_\Omega f_P(\mathbf{y}) d\mathbf{y} = \int_{\partial\Omega} Dh_P(\mathbf{y}) \mathbf{n}(\mathbf{y}) dS. \quad (65)$$

Constant forcing isolates boundary geometry; distance forcing changes the interior load according to clearance; and average-flux forcing chooses a constant load consistent with a prescribed mean boundary flux.

The guidance construction supplies additional directional freedom. Let $\mathbf{n}$ be an approximate outward normal obtained from the signed-distance field and prescribe

$$\mathbf{v}_g(\mathbf{y}) = b(\mathbf{y})\mathbf{n}(\mathbf{y}), \quad \mathbf{y} \in \partial\Omega. \quad (66)$$

Each component is harmonically extended through free space,

$$\Delta(\mathbf{v}_g)_i = 0 \quad \text{in } \Omega, \quad (\mathbf{v}_g)_i|_{\partial\Omega} = b(\mathbf{y})n_i(\mathbf{y}). \quad (67)$$

A smooth sign-preserving map converts its divergence into a negative Poisson load,

$$f_g(\mathbf{y}) = -\frac{1}{\beta} \log(1 + \exp[-\beta \nabla \cdot \mathbf{v}_g(\mathbf{y})]), \quad \beta > 0. \quad (68)$$

The load is normalized to a common mean absolute magnitude when forcing methods are compared. This preserves $f_g < 0$ while retaining spatial information from the guidance divergence. The construction is motivated by the Poisson-safety-function result that guidance fields permit boundary-flux shaping and therefore manipulation of obstacle-adjacent gradients that is unavailable to fixed-gradient distance functions [5].

b) *Selection criterion.*: Figure 2 is a construction diagnostic, not a parameter-selection result: it verifies that a common occupancy and Dirichlet boundary can generate distinct interior fields and derivative profiles. The canonical forcing is selected later by a controlled numerical comparison that holds geometry, normalization, planner, and controller gains fixed; see Section V–B. This separation prevents visual smoothness alone from being mistaken for control suitability.

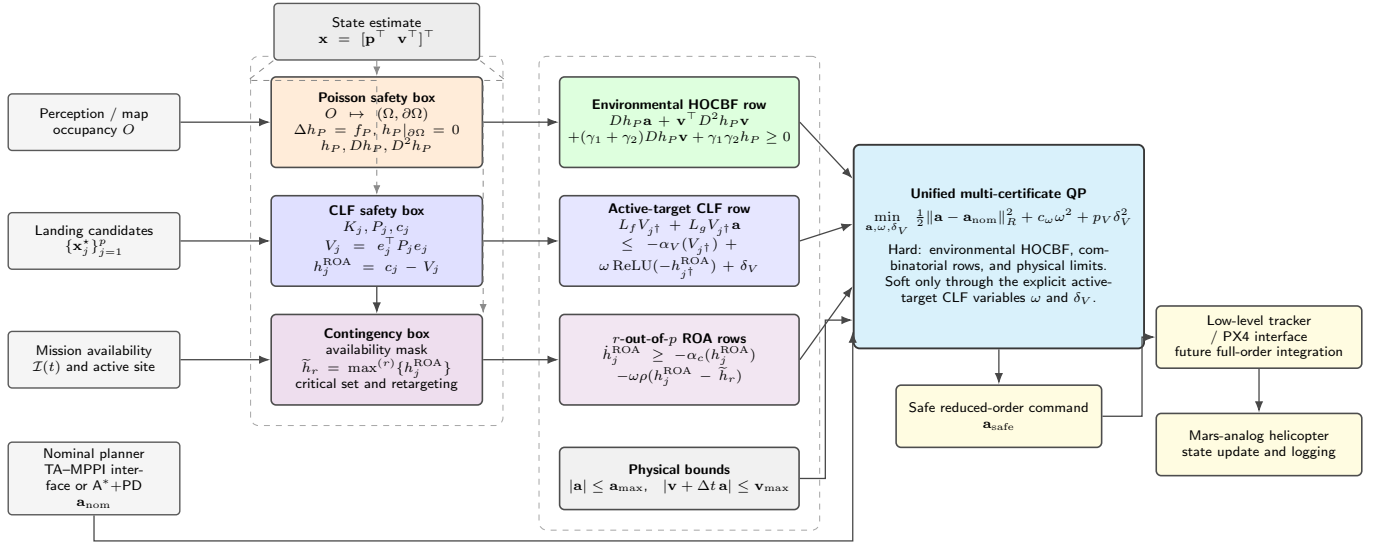


Fig. 1: End-to-end multi-certificate architecture. Perception-derived occupancy is converted into a Poisson safety function; local landing CLFs produce candidate sublevel certificates; the contingency box forms an r -out-of- p order statistic; and all hard and soft rows are assembled in one reduced-order QP. The state is evaluated by every certificate box, while only the filtered acceleration is passed to the lower-level control interface.

TABLE I: Certificate-box interfaces in the proposed architecture.

Module	Input	Output	Role in the unified program
Poisson safety box	Occupancy O , free-space geometry, forcing configuration	h_P, Dh_P, D^2h_P , numerical residuals	Constructs a differentiable representation of environmental safety.
HOCBF box	Current $(\mathbf{p}, \mathbf{v})$ and local Poisson sample	One environmental affine row	Enforces collision avoidance for the relative-degree-two dynamics.
CLF safety box	Model, equilibria $\mathbf{x}_j^*$, input bounds	K_j, P_j, V_j, c_j , and active-target CLF row	Encodes convergence and candidate reduced-order attraction regions.
Contingency box	Available CLF certificates h_j^{ROA}	Pivot $\tilde{h}_r$, critical set, and p smooth rows	Preserves at least r available candidate certificates without enumerating combinations.
Safety-filter box	$\mathbf{a}_{\text{nom}}$, all row bundles, physical bounds	Verified decision $(\mathbf{a}_{\text{safe}}, \omega, \delta_V)$	Computes the minimum-intervention command and checks all residuals explicitly.

For numerical conditioning, the solved field is scaled by a positive constant s_P and shifted by a small control margin h_m ,

$$\bar{h}_P := \frac{h_P}{s_P} - h_m. \quad (69)$$

The derivatives used by the controller are scaled consistently. Trilinear interpolation provides the local sample

$$S_P(\mathbf{p}) = (\bar{h}_P(\mathbf{p}), D\bar{h}_P(\mathbf{p}), D^2\bar{h}_P(\mathbf{p})). \quad (70)$$

On the voxel grid, the finite-difference stencil and Dirichlet values produce a sparse linear system

$$A_h \mathbf{h} = \mathbf{b}. \quad (71)$$

The software exposes sparse-direct, conjugate-gradient, and SOR solvers under the same matrix and right-hand side. Finite differences subsequently yield Dh_P and D^2h_P . The Poisson box records the assembled-system residual

$$r_A = \|A_h \mathbf{h} - \mathbf{b}\|_2 \quad (72)$$

and a reconstructed-Laplacian diagnostic. These quantities must not be conflated: r_A measures solver consistency for the discrete system actually solved, whereas the reconstructed quantity also

contains boundary treatment, discretization, interpolation, and derivative-reconstruction error.

The numerical correspondence in Fig. 3 is the first validation layer: the field certifies the sampled domain actually used by the simulation rather than a separately hand-designed obstacle model. It does not yet establish closed-loop safety; that implication requires the relative-degree-two row in Eq. (74) and its feasibility in the unified program.

D. Step 3: relative-degree-two environmental HOCBF

For the reduced-order model, the Poisson field depends on position but the control input is acceleration. Define

$$\psi_0 = \bar{h}_P(\mathbf{p}), \quad \psi_1 = D\bar{h}_P(\mathbf{p})\mathbf{v} + \gamma_1 \bar{h}_P(\mathbf{p}). \quad (73)$$

Enforcing $\dot{\psi}_1 + \gamma_2 \psi_1 \geq 0$ yields

$$D\bar{h}_P(\mathbf{p})\mathbf{a} \geq -\mathbf{v}^\top D^2\bar{h}_P(\mathbf{p})\mathbf{v} - (\gamma_1 + \gamma_2)D\bar{h}_P(\mathbf{p})\mathbf{v} - \gamma_1\gamma_2\bar{h}_P(\mathbf{p}). \quad (74)$$

This is the exact acceleration-level row implemented by the environmental box. It is affine in $\mathbf{a}$, while the Hessian term accounts for curvature of the numerical safety field along the current velocity direction. The constraint is treated as

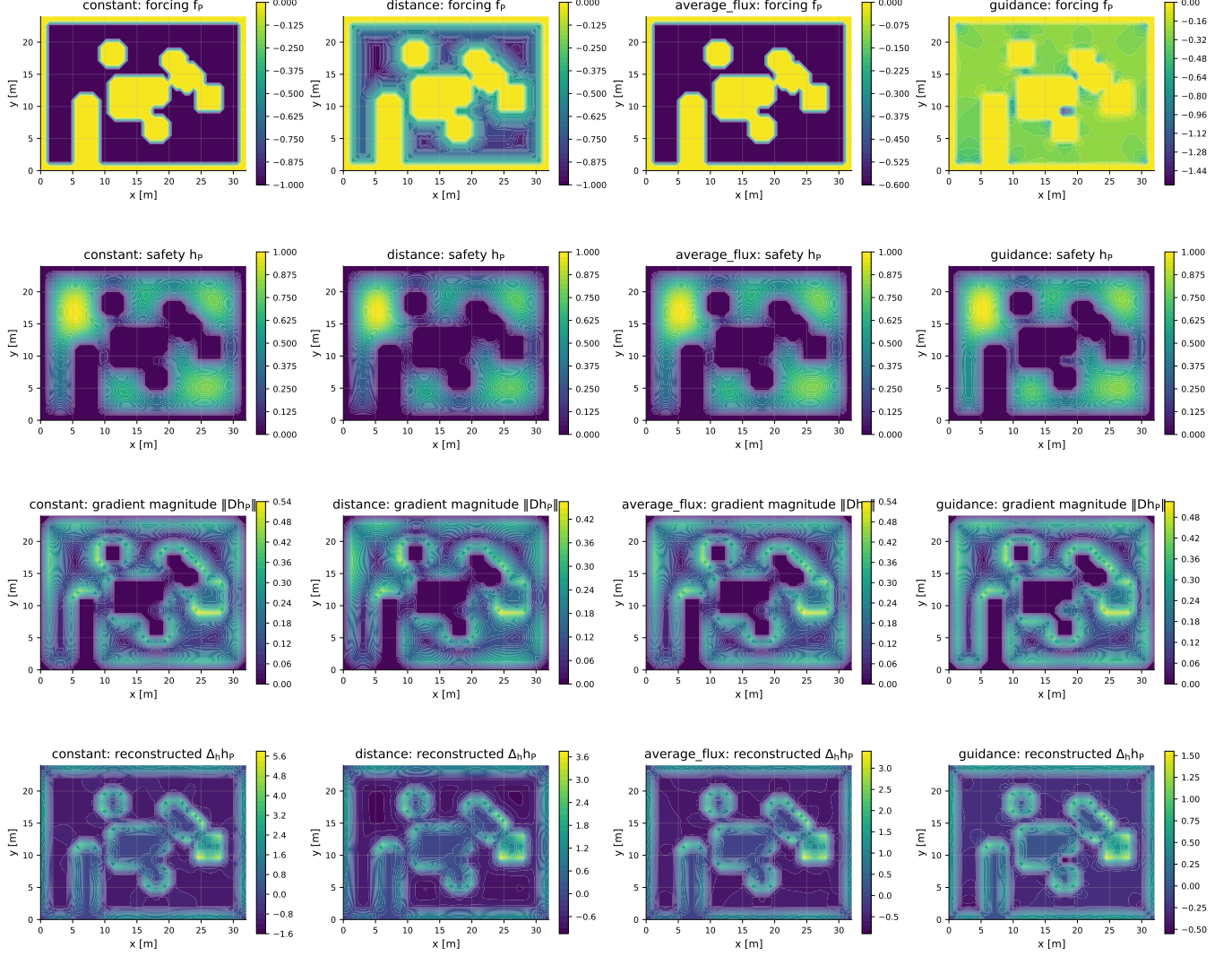


Fig. 2: Forcing-field construction on a common occupancy map. Columns correspond to constant, distance, average-flux, and guidance loading; rows expose the forcing, solved safety field, gradient magnitude, and derivative structure subsequently sampled by the HOCBF. Since Ω and $\partial\Omega$ are fixed, the obstacle set is identical in every column; only the interior geometry of h_P changes.

hard in the unified filter. Under the regularity, initial-set, and locally Lipschitz controller conditions of the HOCBF result, its satisfaction renders the associated higher-order safe set forward invariant [6].

E. Step 4: landing-site CLFs and input-feasible sublevel sets

For the double-integrator model, the linearization about every landing equilibrium is the same pair (A, B) in (51). The implementation first constructs a stabilizing gain K_j from a continuous-time LQR problem. Equivalently, if S_j solves the algebraic Riccati equation, then

$$K_j = R_j^{-1} B^\top S_j, \quad A_{cl,j} = A - B K_j. \quad (75)$$

A separate positive-definite matrix P_j is obtained from

$$A_{cl,j}^\top P_j + P_j A_{cl,j} = -Q_{V,j}, \quad (76)$$

and the landing CLF is

$$V_j(\mathbf{x}) = \mathbf{e}_j^\top P_j \mathbf{e}_j, \quad \mathbf{e}_j := \mathbf{x} - \mathbf{x}_j^*. \quad (77)$$

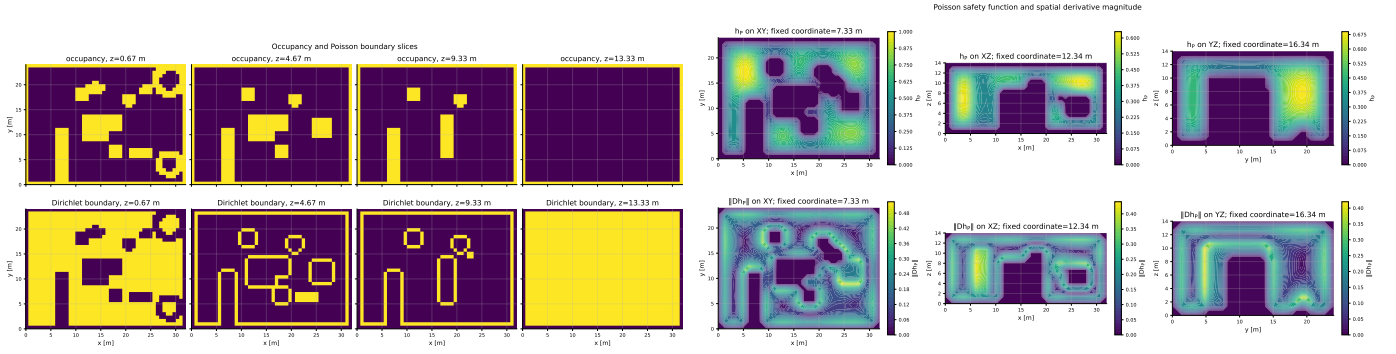
Its differential quantities are available analytically,

$$DV_j = 2\mathbf{e}_j^\top P_j, \quad L_f V_j = DV_j \begin{bmatrix} \mathbf{v} \\ 0 \end{bmatrix}, \quad L_g V_j = DV_j B. \quad (78)$$

The active-target decrease function is presently linear, $\alpha_V(V_j) = k_V V_j$, although the software interface permits alternative class- $\mathcal{K}$ choices.

The sublevel threshold is not selected solely for visual convenience. Let $\mathbf{k}_{j,i}$ denote row i of K_j . Over the ellipsoid $\mathbf{e}_j^\top P_j \mathbf{e}_j \leq c$, the maximum magnitude of the corresponding linear feedback component is

$$\max_{\mathbf{e}_j^\top P_j \mathbf{e}_j \leq c} |\mathbf{k}_{j,i} \mathbf{e}_j| = \sqrt{c \mathbf{k}_{j,i} P_j^{-1} \mathbf{k}_{j,i}^\top}. \quad (79)$$



(a) Inflated occupancy and extracted Dirichlet boundary on representative altitude slices. (b) Representative planes of the solved Poisson safety function and its spatial variation.

Fig. 3: Procedural environmental construction. Panel (a) checks that the inflated occupancy and extracted frontier define the same obstacle geometry used by the solver; panel (b) shows the resulting positive free-space field on representative planes. Together they validate the map-to-boundary-to-field data path, while the HOCBF guarantee still depends on the derivative row derived below.

Consequently, an input-feasible level is obtained from

$$c_{j,\max}^u = \min_i \frac{q_{\max,i}^2}{\mathbf{k}_{j,i} P_j^{-1} \mathbf{k}_{j,i}^\top}, \quad (80a)$$

$$c_j = \eta_{\text{ROA}} c_{j,\max}^u, \quad 0 < \eta_{\text{ROA}} < 1. \quad (80b)$$

The resulting certificate is $h_j^{\text{ROA}} = c_j - V_j$. For the current linear reduced model, Eq. (80) certifies compatibility with the componentwise LQR input bounds. It does not imply that every point in the ellipsoid is collision-free; environmental safety is imposed independently by the Poisson HOCBF.

Figure 4 validates implementation of the certificate maps and order statistic, but it is not by itself a proof that a shown spatial ellipse is a global attraction region. The formal statement is restricted to sublevel sets contained in the domain where the CLF decrease, input feasibility, and environmental constraints are jointly valid.

F. Step 5: available-site combinatorial contingency

At every control update, unavailable candidates are removed before the order statistic is computed. The contingency box evaluates

$$\tilde{h}_r^{\mathcal{I}} = \max^{(r)} \{h_j^{\text{ROA}}\}_{j \in \mathcal{I}(t)}, \quad (81)$$

its certified count, and the set of certificates equal to the pivot within a numerical tolerance. For the current implementation,

$$\alpha_c(h) = k_c h, \quad \rho(s) = k_\rho s^2. \quad (82)$$

One affine row is then generated for every available candidate,

$$\begin{aligned} \dot{h}_j^{\text{ROA}}(\mathbf{x}, \mathbf{a}) &\geq -k_c h_j^{\text{ROA}}(\mathbf{x}) \\ &\quad - \omega k_\rho \left(h_j^{\text{ROA}}(\mathbf{x}) - \tilde{h}_r^{\mathcal{I}}(\mathbf{x}, t) \right)^2, \end{aligned} \quad (83)$$

where the same $\omega \geq 0$ is used in every row. Because $\dot{h}_j^{\text{ROA}} = -\dot{V}_j$, these inequalities are affine in acceleration. The construction requires only p smooth contingency rows, not one row for each of the $\binom{p}{r}$ combinations [9], [10].

The selected CLF uses the theorem-compatible shared auxiliary variable,

$$\begin{aligned} L_f V_{j^\dagger} + L_g V_{j^\dagger} \mathbf{a} &\leq -\alpha_V(V_{j^\dagger}) \\ &\quad + \omega \text{ReLU}(-h_{j^\dagger}^{\text{ROA}}) + \delta_V. \end{aligned} \quad (84)$$

Inside the selected candidate sublevel set, $\text{ReLU}(-h_{j^\dagger}^{\text{ROA}}) = 0$, and the shared- ω term vanishes. The additional nonnegative variable δ_V is an explicit CLF slack used to prevent a stability objective from making the hard safety program infeasible. Neither the environmental HOCBF nor the combinatorial rows receive this generic slack.

G. Step 6: unified multi-certificate quadratic program

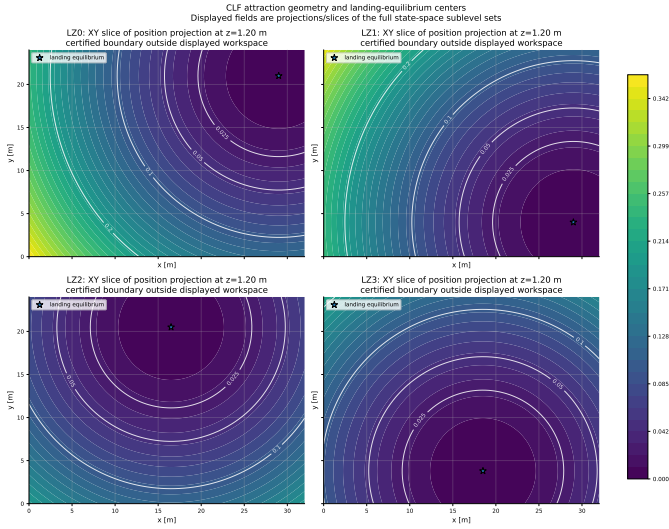
The augmented decision is

$$\mathbf{z} := \begin{bmatrix} \mathbf{a} \\ \omega \\ \delta_V \end{bmatrix}. \quad (85)$$

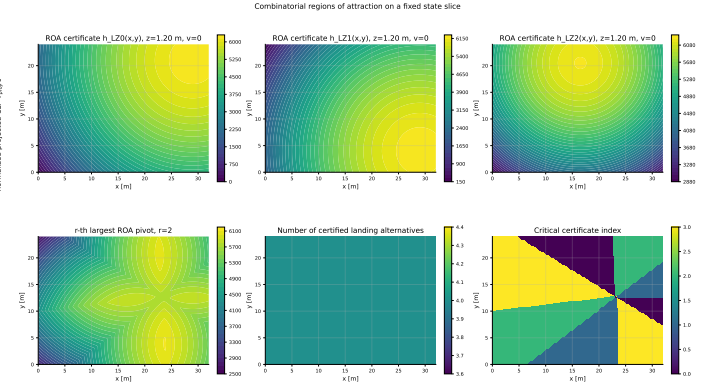
At each control update, the complete filter is

$$\begin{aligned} \mathbf{z}^* &= \arg \min_{\mathbf{z}} \frac{1}{2} \|\mathbf{a} - \mathbf{a}_{\text{nom}}\|_R^2 + c_\omega \omega^2 + p_V \delta_V^2 \\ \text{s.t. } & D\bar{h}_P \mathbf{a} + \mathbf{v}^\top D^2 \bar{h}_P \mathbf{v} \\ & \quad + (\gamma_1 + \gamma_2) D\bar{h}_P \mathbf{v} + \gamma_1 \gamma_2 \bar{h}_P \geq 0, \\ & L_f V_{j^\dagger} + L_g V_{j^\dagger} \mathbf{a} + \alpha_V(V_{j^\dagger}) \\ & \quad - \omega \text{ReLU}(-h_{j^\dagger}^{\text{ROA}}) - \delta_V \leq 0, \\ & \dot{h}_j^{\text{ROA}} + \alpha_c(h_j^{\text{ROA}}) \\ & \quad + \omega \rho(h_j^{\text{ROA}} - \tilde{h}_r^{\mathcal{I}}) \geq 0, \\ & \quad j \in \mathcal{I}(t), \\ & -\mathbf{a}_{\max} \preceq \mathbf{a} \preceq \mathbf{a}_{\max}, \\ & -\mathbf{v}_{\max} \preceq \mathbf{v} + \Delta t \mathbf{a} \preceq \mathbf{v}_{\max}, \\ & \omega \geq 0, \quad \delta_V \geq 0. \end{aligned} \quad (86)$$

The safe acceleration $\mathbf{a}_{\text{safe}}$ is the acceleration block of $\mathbf{z}^*$; the remaining blocks are the shared contingency variable ω^* and the explicit CLF slack δ_V^* . The objective is strictly convex for positive weights. Every constraint can be written in the canonical form (60); therefore, the reduced-order online problem is a convex QP [18]. The implementation verifies $Az - b$ and the componentwise bounds after every solve rather than accepting solver status as a certificate of feasibility.



(a) Quadratic CLF contours and projected candidate sublevel regions centered at the landing equilibria.



(b) Spatial slices of candidate certificates, certified count, critical identity, and the r -th-largest pivot.

Fig. 4: CLF and contingency objects used by the filter. Panel (a) verifies that each quadratic form is centered at its landing equilibrium; panel (b) verifies the candidate-margin, certified-count, critical-index, and r -th-pivot computations on a common spatial slice. The displayed sets are fixed-velocity slices or projections of the full six-dimensional position–velocity sublevel sets.

Algorithm 1 One multi-certificate control update

Require: State $\mathbf{x}_k$, occupancy/Poisson snapshot, available set

- 1: Evaluate all V_j , h_j^{ROA} , and availability flags
- 2: **if** $j_k^\dagger \notin \mathcal{I}_k$ **then**
- 3: Select the available certified site with maximum ROA margin
- 4: **end if**
- 5: Compute $\mathbf{a}_{\text{nom},k}$ from the nominal planner
- 6: Query $(\bar{h}_P, D\bar{h}_P, D^2\bar{h}_P)$ at $\mathbf{p}_k$
- 7: Build the HOCBF, active CLF, contingency, and physical-bound rows
- 8: **if** $|\mathcal{I}_k| < r$ or $\tilde{h}_r^{\mathcal{I}} < 0$ **then**
- 9: Report loss of the requested contingency property
- 10: **else**
- 11: Solve (86) and verify all residuals
- 12: Apply $\mathbf{a}_{\text{safe},k}$ under zero-order hold
- 13: **end if**

H. Step 7: runtime procedure and closed-loop interpretation

The Poisson field is precomputed once for the static three-dimensional study. In the image and streaming modes, field construction is placed in a versioned latest-only worker, while CLF, contingency, and QP evaluations remain in the control loop. This division follows the layered principle that slow perception-derived safety synthesis and fast pointwise filtering should have separate timing contracts [8].

Proposition 1 (Reduced-order certificate implication). *Assume that the initial condition lies in the HOCBF admissible set and in $\tilde{\mathcal{R}}_r^{\mathcal{I}}(0)$, the Poisson and CLF regularity conditions hold, the available set is piecewise constant between declared target events, and the solution of (86) is locally Lipschitz and feasible. Then the hard environmental HOCBF and combinatorial rows*

render their respective sets forward invariant on each interval between target events. If the active site remains available, its candidate region is valid, and $\delta_V = 0$, then the active CLF satisfies its nominal decrease condition and the reduced-order state converges to the selected landing equilibrium. If $\delta_V > 0$, the hard safety and contingency implications remain, but asymptotic convergence is not implied by that sample.

Proposition 1 follows by combining the HOCBF forward-invariance result, the CLF stabilization lemma, and the combinatorial stabilization construction reviewed in Section II and Appendix A. It applies only to the stated reduced-order model and certificate assumptions; it is not yet a full-order PX4 or hardware guarantee.

V. EXPERIMENTAL SETUP

The present campaign is a deterministic reduced-order validation of the objects introduced in Section IV. Its role is not to replace the final statistical and hardware Results section. Instead, it establishes that the geometry, Poisson field, derivatives, certificate rows, target manager, and unified QP are mutually consistent before deployment on PX4 or hardware. Every figure in this section is therefore tied to a specific methodological question: whether the environment is nontrivial, which forcing and linear solver should be retained, which certificate gains admit safe progress under sampled integration, whether the landing test is actually satisfied, and what is—and is not—demonstrated by the contingency traces.

A. Mars-analog world, nominal layer, and touchdown test

The workspace measures $32 \text{ m} \times 24 \text{ m} \times 14 \text{ m}$ and is sampled on a $48 \times 36 \times 22$ voxel grid. Fourteen analytic obstacles are rasterized and inflated by 0.28 m before constructing the Poisson domain. The scene contains escarpments, a central mesa, an elevated arch, high spires, suspended rocks and

shelves, low terrain, and annular crater rims. The initial state is

$$\mathbf{x}_0 = [2, 2, 12, 0, 0, 0]^\top, \quad (87)$$

and the four landing equilibria are centered at

$$\begin{aligned} \mathbf{p}_0^* &= [29, 21, 1.2]^\top, & \mathbf{p}_1^* &= [29, 4, 1.2]^\top, \\ \mathbf{p}_2^* &= [16.5, 20.5, 1.2]^\top, & \mathbf{p}_3^* &= [18.5, 3.8, 1.2]^\top. \end{aligned} \quad (88)$$

LZ0 is the primary target and the required contingency count is $r = 2$. The straight segment from (87) to LZ0 intersects occupied voxels in 42.8% of its sampled checks. Hence the scenario cannot be explained by an unobstructed straight descent. The baseline nominal layer uses a clearance-weighted three-dimensional A* path and the PD law in (61).

A landing is declared only when both

$$\|\mathbf{p} - \mathbf{p}_{j^\dagger}^*\|_2 \leq \varepsilon_p, \quad \|\mathbf{v}\|_2 \leq \varepsilon_v, \quad (89)$$

with $\varepsilon_p = 0.55$ m and $\varepsilon_v = 0.35$ m s⁻¹. Visual proximity or a final point above the landing zone is therefore insufficient.

Figure 5 supports two distinct statements. First, the obstacle map invalidates the straight mission reference and therefore exercises the planner/environment interface. Second, the filtered baseline reaches LZ0 with terminal position error 0.227 m and terminal speed 0.336 m s⁻¹, while retaining minimum obstacle clearance 1.333 m, minimum Poisson value 0.1368, and minimum HOCBF residual 4.98×10^{-2} . These observations verify reduced-order collision-free execution and the touchdown test. They do not, by themselves, isolate the contribution of the HOCBF because the A* nominal path is already clearance aware; a direct-nominal ablation remains a separate final Results experiment.

B. Poisson forcing and linear-solver calibration

The forcing comparison holds occupancy, boundary values, normalization, planner, and all controller gains fixed while changing only $f_P \in \{f_c, f_d, f_{af}, f_g\}$ from Eqs. (64)–(68). The solver comparison then holds the selected guidance field, sparse operator, and right-hand side fixed while changing only the numerical method. This controlled design prevents changes in geometry or feedback gains from being misattributed to the PDE construction.

All forcing variants complete the deterministic baseline landing, so the selection is made from controller-relevant margin rather than binary success. Guidance produces the largest retained minimum Poisson value, $\min h_P = 0.1368$, and the largest minimum HOCBF residual, $\min r_{\text{HOCBF}} = 4.98 \times 10^{-2}$. Constant and average-flux forcing yield $\min h_P = 0.0695$ and $\min r_{\text{HOCBF}} = 4.30 \times 10^{-3}$, while distance forcing reaches $\min h_P = 0.0466$ with an essentially active HOCBF row. Guidance requires 1.180 s to construct in the current implementation, compared with 0.296 s to 0.913 s for the alternatives. Because the static field is generated before rollout, the canonical experiment retains guidance and prioritizes derivative margin over offline construction time. This is a map- and normalization-specific conclusion, not a universal ranking of forcing functions.

a) *Solver selection.*: The solver study is shown separately in Fig. 7 so that its time–accuracy trade is readable at column scale rather than compressed into the forcing montage.

For the selected guidance field, sparse direct requires 0.497 s and attains algebraic residual 1.14×10^{-14} . Conjugate gradient reduces the solve to 0.300 s while maintaining relative field error 1.93×10^{-8} and residual 8.14×10^{-8} . SOR is fastest at 0.070 s but increases the field error to 1.03×10^{-6} and the residual to 1.10×10^{-5} . Conjugate gradient is therefore retained as the accuracy–time compromise for this mesh. The common reconstructed-Laplacian error is not used to rank solvers, because it also contains boundary, discretization, and derivative- reconstruction effects shared by all three methods.

C. Certificate-gain and candidate-region sweeps

The HOCBF study scales both gains as $\gamma_i = s_\gamma \gamma_i^0$ and uses a deliberately direct nominal reference on the same obstacle world. This makes the safety filter, rather than A*, responsible for reconciling the unsafe task command. The tested scales are

$$s_\gamma \in \{0.45, 0.65, 0.85, 1.10, 1.40, 1.80, 2.40\}. \quad (90)$$

The active-target CLF sweep uses $k_V \in \{0.012, 0.018, 0.025, 0.03, 0.045, 0.065, 0.09\}$, and the candidate- region study uses $\eta_{\text{ROA}} \in \{0.55, 0.70, 0.85, 0.96\}$.

Figure 8 identifies a successful HOCBF interval $0.65 \leq s_\gamma \leq 1.40$. The most conservative case terminates on the residual monitor before landing, while $s_\gamma \in \{1.80, 2.40\}$ reaches the sampled-data collision guard. The retained value $s_\gamma = 1.10$ lies in the interior of the successful interval. This result also shows why continuous-time constraint satisfaction must be accompanied by a finite-step collision diagnostic.

All retained CLF-gain runs land, but larger k_V increases intervention and moves the minimum Poisson value toward the boundary. The intermediate value $k_V = 0.03$ is retained. Changing η_{ROA} primarily rescales the pivot; all four candidates remain certified and the shared auxiliary variable satisfies $\omega = 0$ throughout the baseline. Consequently, the ROA sweep verifies numerical consistency of the candidate certificates, but it does not demonstrate active trajectory modification by the combinatorial rows. That limitation is carried explicitly into the claim table below.

D. Primary landing, diversion, and contingency exhaustion

The final simulation block examines how the same closed-loop architecture behaves when the availability of the landing sites changes during flight. To keep the comparison interpretable, the world, occupancy grid, Poisson field, reduced-order dynamics, planner gains, certificate parameters, QP weights, and touchdown tolerances are identical in all three runs. Only the availability schedule is changed. The three cases therefore answer three separate questions rather than forming a single increasingly difficult trajectory:

- 1) *Primary landing*: can the vehicle traverse the obstacle-rich world and satisfy the touchdown conditions at LZ0 when no target fails?

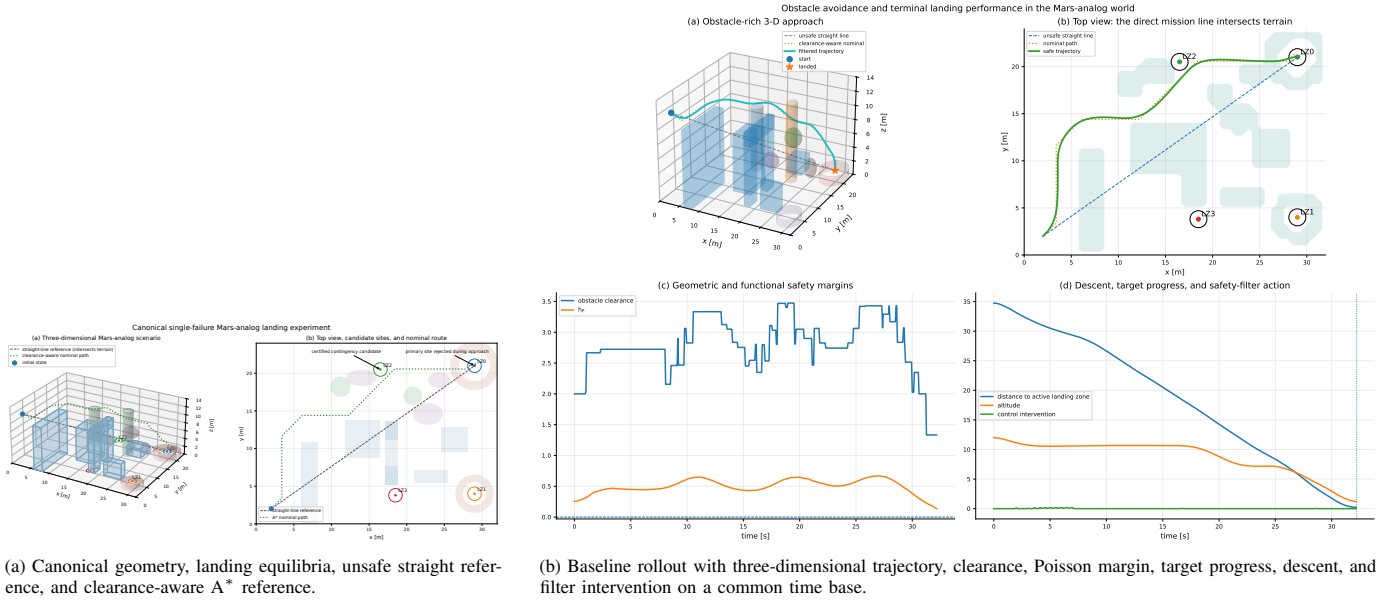


Fig. 5: World definition and primary-landing verification. Panel (a) establishes that the geometric task is nontrivial; panel (b) evaluates whether the nominal-plus-filter stack traverses a free corridor and satisfies the terminal test rather than merely approaching LZ0.

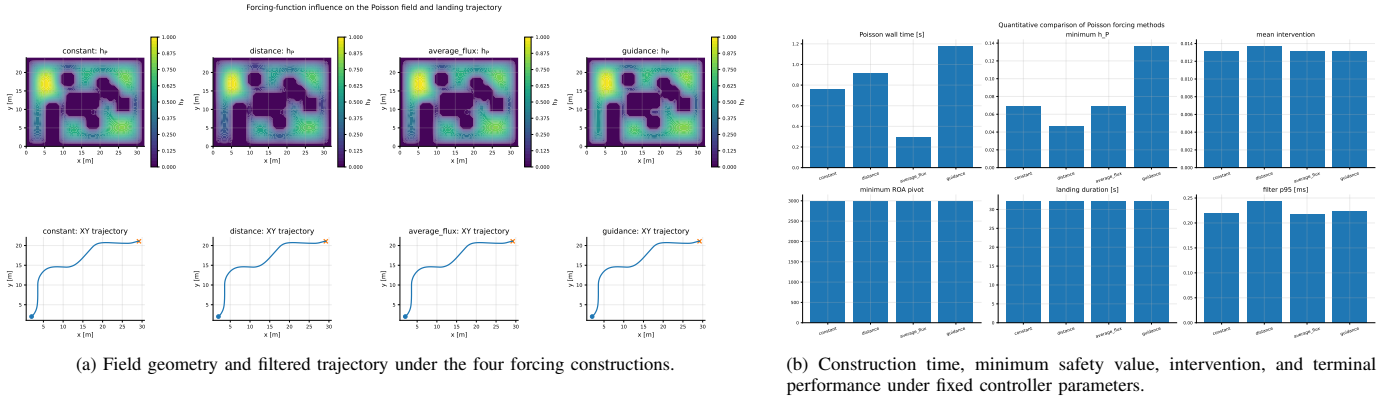


Fig. 6: Controlled forcing-field study. The occupancy and Dirichlet boundary are identical in every run; the observed differences therefore arise from the interior field and derivatives supplied to the HOCBF.

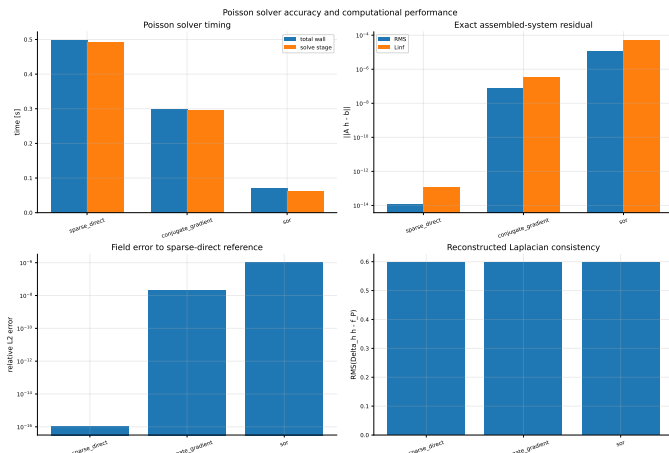
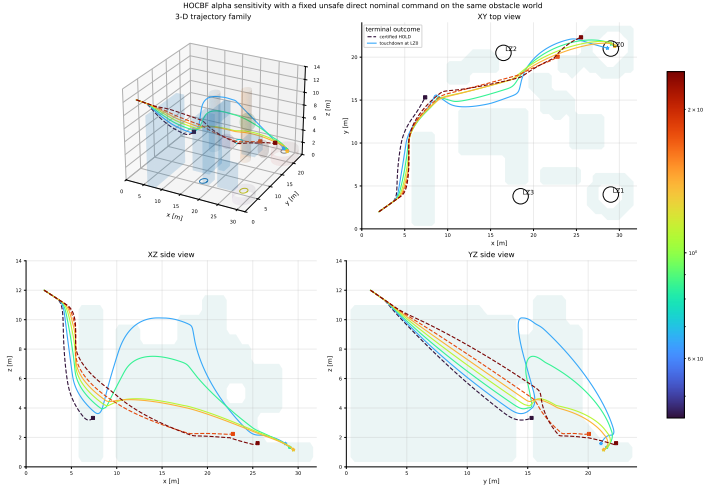


Fig. 7: Poisson solver comparison for a common sparse operator and right-hand side. Algebraic residual and relative field error determine numerical fidelity; the common reconstructed-Laplacian error is treated as a discretization/derivative diagnostic rather than a solver ranking.

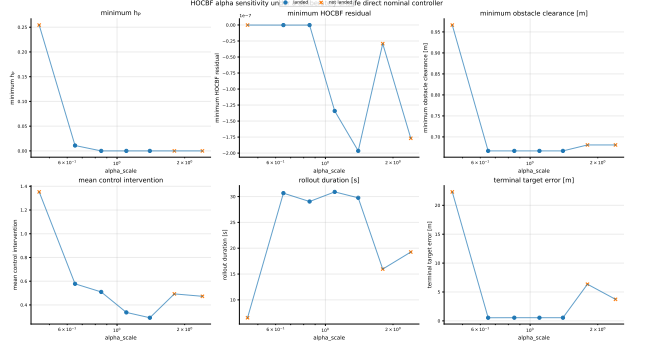
- 2) *Single-site diversion*: after LZ0 is rejected, can the target manager select another available candidate and can the same safety filter complete the new landing task?
- 3) *Sequential-site exhaustion*: when targets are removed one after another, does the runtime detect the instant at which the requested $r = 2$ contingency requirement can no longer be supported?

This separation is important. The first two cases end in a landing and can be checked against Eq. (89). The third case is a logic and monitoring stress test; it is not expected to satisfy a landing condition.

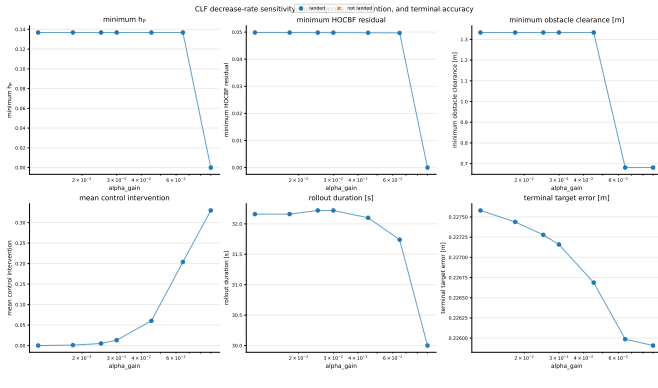
a) *What the scenario-comparison plot establishes.:* The upper part of Fig. 9 is constructed from the logged (x, y) state history, the active-target label at each sample, the projected occupancy map, and the recorded target-switch events. The baseline trace is a single target-labelled curve ending at LZ0. The one-failure trace contains a blue segment toward LZ0 followed by an orange segment toward LZ2; the change of color is therefore the geometric record of the retargeting event,



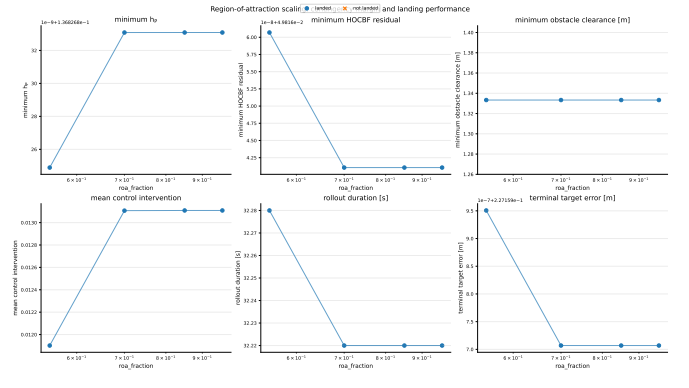
(a) Trajectory family for the HOCBF-gain sweep on one fixed obstacle world.



(b) Safety, intervention, terminal, and timing metrics versus $\alpha\gamma$.



(c) Active-target CLF decrease-gain sweep.



(d) Input-feasible candidate-sublevel scaling sweep.

Fig. 8: Certificate-parameter studies. The HOCBF sweep identifies a sampled-data safety-progress regime; the CLF sweep calibrates convergence demand; and the ROA-fraction sweep tests certificate scaling and whether the current scenario actually activates the combinatorial rows.

not a separately drawn reference curve. The sequential trace contains three target-labelled segments and shows how the commanded mission changes after successive site removals. This panel contributes a mission-level view: it makes the switch visible in the same coordinates as the obstacles and candidate sites. It does not, on its own, prove touchdown, certificate satisfaction, or a stabilized HOLD state.

b) How touchdown is verified.: The two lower panels of Fig. 9 are generated from the final logged position and velocity, together with the prescribed tolerances ε_p and ε_v . In the baseline case, the vehicle reaches LZ0 at $t = 32.22$ s with

$$\|\mathbf{p}(T) - \mathbf{p}_0^*\|_2 = 0.227 \text{ m}, \quad \|\mathbf{v}(T)\|_2 = 0.336 \text{ m s}^{-1}. \quad (91)$$

Both values lie inside the corresponding bounds in Eq. (89). The lower-left panel is therefore evidence of a completed reduced-order landing, rather than merely a trajectory that passes above the target.

For the one-failure run, LZ0 is rejected at $t = 24.90$ s. The target manager then selects LZ2, and the vehicle reaches that equilibrium at $t = 33.90$ s with

$$\|\mathbf{p}(T) - \mathbf{p}_2^*\|_2 = 0.026 \text{ m}, \quad \|\mathbf{v}(T)\|_2 = 0.328 \text{ m s}^{-1}. \quad (92)$$

The lower-right panel contributes the terminal part of the contingency argument: after the mission target changes, the

state converges to the new landing equilibrium and satisfies the same landing definition used in the baseline. This avoids weakening the success criterion after a failure.

c) Reading the one-failure timeline.: Figure 10a(a) repeats the spatial retargeting trace, but now places the failure and switch markers directly on the trajectory. Panel (d) supplies the corresponding temporal explanation: the distance to LZ0 decreases until the rejection time, whereas the distance to LZ2 subsequently decreases to zero. Together these two panels show that the vehicle does not simply continue along the original path after the target is removed.

Panel (b) is obtained by evaluating every quadratic certificate $h_j^{\text{ROA}} = c_j - V_j$ along the same state trajectory. The purple curve is the second-largest value because $r = 2$. Its positivity throughout the run shows that the state remains inside at least two of the candidate sublevel sets used by the implementation. Panel (c) adds the discrete mission information: the available-site count falls from four to three when LZ0 is rejected, while the required count remains two. Thus the switch is permitted without violating the configured availability requirement. These plots support consistent certificate evaluation and retargeting. Because the margins remain far above zero, however, they do not show that the combinatorial rows had to actively reshape

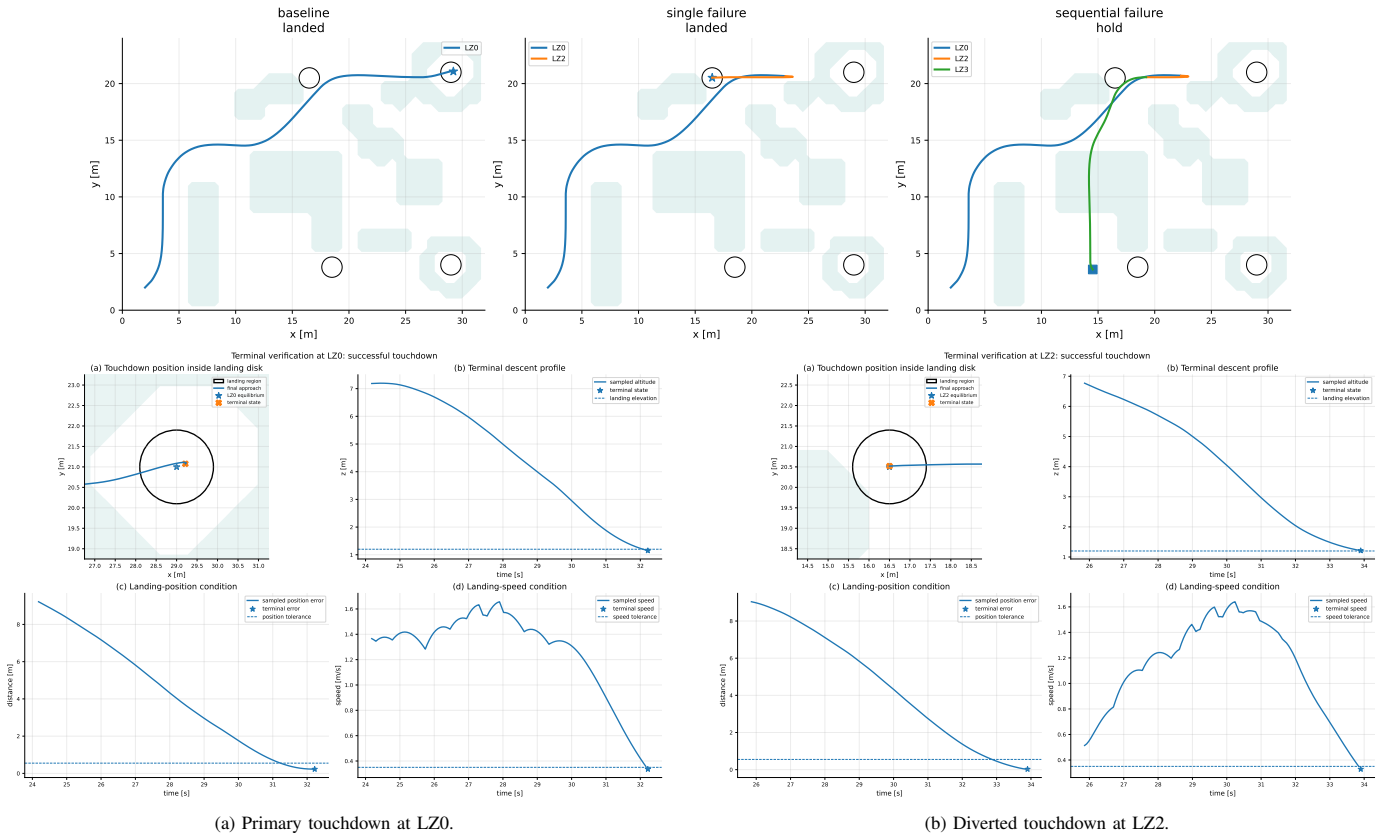


Fig. 9: Scenario comparison and terminal verification. The upper row shows the horizontal path in the same obstacle projection for the baseline, one-failure, and sequential-failure runs. Trajectory color identifies the active landing target. The lower panels evaluate the two successful terminal states against the position and speed tolerances, replacing a visual “near the site” judgment with the quantitative touchdown test in Eq. (89).

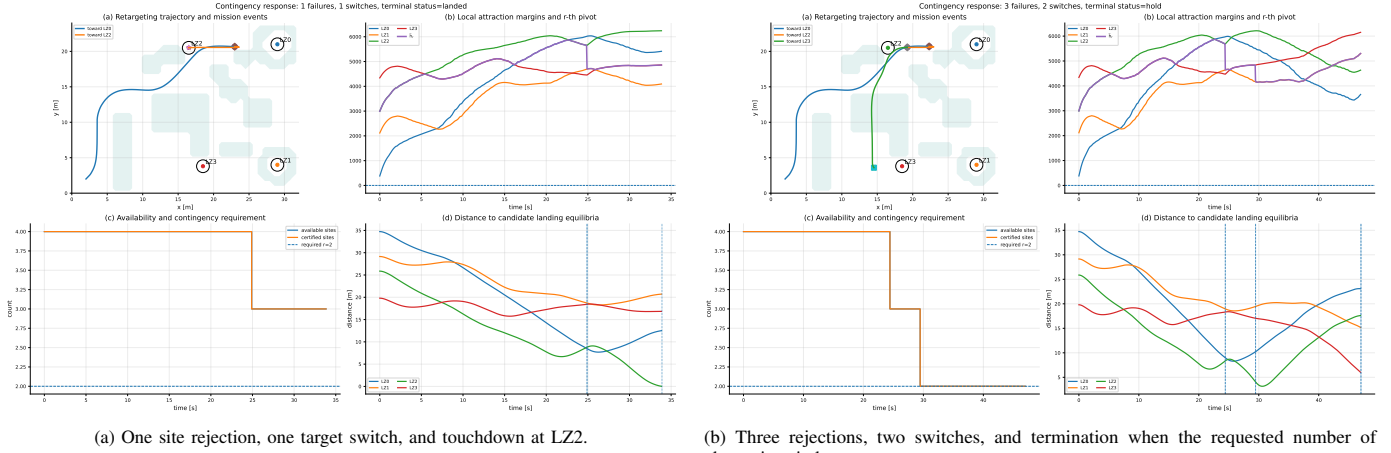


Fig. 10: Contingency timelines generated from the state history, availability log, target-event log, and per-target CLF certificates. In each panel, (a) locates the retargeting events in the workspace, (b) reports the candidate margins and the r -th pivot, (c) compares the available/certified counts with the required value $r = 2$, and (d) shows distance to every landing equilibrium.

the trajectory.

d) Reading the sequential-failure timeline.: Figure 10b is generated in the same way, but with three scheduled site removals. The target-labelled path in panel (a) records the sequence $LZ0 \rightarrow LZ2 \rightarrow LZ3$. The vertical event lines in panel (d) align these switches with the distances to all four equilibria. Panel (b) shows that removing a site changes which

certificate determines the r -th pivot even though the continuous state is unchanged at the event instant. Panel (c) then gives the mission-level reason for termination: after the final rejection the event log contains only one available landing site, fewer than the required $r = 2$.

The final removal occurs at the terminal sample, so the staircase in panel (c) primarily displays the pre-termination sampled

counts; the event log is the authoritative record of the final transition from two available sites to one. The runtime stops at $t = 47.04\text{ s}$ with $\|\mathbf{v}(T)\|_2 = 1.428\text{ m s}^{-1}$. Consequently, this experiment validates detection of contingency exhaustion and correct event handling. It does not validate braking to zero velocity, hovering, or a physically stabilized HOLD maneuver.

e) What the synchronized certificate histories add.:

Figure 11 connects the mission-level plots to the individual rows of the optimization problem. The upper-left panel is the sampled Poisson value $h_P(\mathbf{p}(t))$; it remains positive and approaches its smallest values during final descent. The upper-right panel shows the corresponding HOCBF residual. Its nonnegative evolution, up to numerical tolerance, verifies that the hard environmental row is satisfied along the logged rollout rather than only at the terminal state.

The second row concerns stabilization. The active-target Lyapunov function falls as the vehicle approaches LZ0, changes value when the identity of the active equilibrium changes, and then decreases toward zero for LZ2. This jump is expected: the state is continuous, but the function being evaluated changes from V_0 to V_2 . The adjacent CLF-residual panel indicates when the requested decrease condition is demanding and when the relaxation is no longer needed near touchdown. These panels contribute evidence that the switch is propagated into the stabilization certificate, not only into the plot legend or mission manager.

The third row reports all four candidate values and their r -th pivot. Every candidate remains well above the zero boundary in this experiment. The bottom-left panel explains the control consequence: the norm $\|\mathbf{a}_{\text{safe}} - \mathbf{a}_{\text{nom}}\|$ is bounded, the CLF slack is small, and the shared combinatorial auxiliary satisfies $\omega = 0$. Thus the one-failure result demonstrates successful monitored retargeting, but not a case in which the combinatorial constraints actively intervene. Finally, the bottom-right panel places the computation in context: the pointwise multi-certificate filter remains below approximately 0.4 ms in this deterministic run. This timing refers to the QP/filter update with a precomputed field; it does not include rebuilding the three-dimensional Poisson field.

Taken together, the figures in this subsection support three progressively stronger but distinct conclusions: the baseline satisfies the formal landing criterion; a single exogenous site rejection is followed by a consistent certificate update, target switch, and successful diverted landing; and a sequence of rejections is detected when it exhausts the configured availability requirement. They do not yet establish full-order PX4 landing, robustness to state-estimation error, active preservation of a near-boundary r -out-of- p set, or a stabilized physical HOLD after contingency exhaustion.

Table II records the inference supported by each study and the corresponding limitation. This evidence-to-claim map prevents plots from being treated as decoration or from supporting stronger statements than the data permit.

TABLE II: Evidence-to-claim map for the reduced-order campaign.

Supported inference and evidence	Scope limitation
Geometric task, Fig. 5. The straight reference intersects occupancy; the nominal/filter stack uses a three-dimensional free corridor and completes touchdown.	Does not isolate the HOCBF from the clearance-aware A* nominal policy.
Forcing, Fig. 6. Guidance gives the largest sampled h_P and HOCBF margins under controlled normalization.	No map-independent optimal-forcing or online real-time construction claim.
Linear solver, Fig. 7. Conjugate gradient gives the retained accuracy–time compromise on the canonical grid.	No scaling claim across grids, processors, or hardware.
HOCBF gain, Fig. 8(a)–(b). $s_\gamma = 1.10$ lies inside the observed landing regime and away from the sampled-data failure cases.	No arbitrary intersample-safety claim without the collision guard.
CLF/pivot, Fig. 8(c)–(d) and Fig. 11. The quadratic CLFs, candidate margins, and pivot are evaluated consistently.	No active nontrivial r -out-of- p preservation claim; $\omega = 0$ here.
Diversification, Figs. 9 and 10. LZ0 is removed, LZ2 is selected, and the diverted touchdown conditions are satisfied.	No full-order PX4/hardware or state-estimation robustness claim.
Repeated failures, Fig. 10(b). The runtime detects loss of the configured $r = 2$ availability requirement.	Not a stabilized physical HOLD; termination occurs with nonzero velocity.

E. Canonical parameters and reproducibility

Table III lists the retained configuration. The reduced-order QP is solved by a warm-started Hildreth dual-coordinate method, after which every affine and bound residual is recomputed explicitly.

TABLE III: Retained canonical simulation parameters.

Parameter	Retained value
World / grid	$32 \times 24 \times 14\text{ m}$; $48 \times 36 \times 22$; 14 inflated obstacles
Sampling	$\Delta t = 0.06\text{ s}$; at most 1200 steps
Physical bounds	$a_{\text{max}} = 5.5\text{ m s}^{-2}$; $v_{\text{max}} = 3.0\text{ m s}^{-1}$
Nominal layer	A*+PD; $k_p = 0.72$, $k_v = 1.08$, look-ahead 2.2 m
Poisson box	Guidance, $\beta = 8$; conjugate gradient, tolerance 10^{-7}
HOCBF	$\gamma_1 = \gamma_2 = 1.1$; $h_m = 0.003$
Landing CLF	$\alpha_V(V) = 0.03V$; $\eta_{\text{ROA}} = 0.96$
Contingency	$r = 2$; $\alpha_c(h) = 0.16h$; $\rho(s) = s^2$
Unified QP	Hildreth; $c_\omega = 30$; $p_V = 10^5$

For all four equilibria, the constructed reduced-model artifacts share

$$c_j = 6246.20, \quad c_{j,\text{max}}^u = 6506.46, \quad (93)$$

with Lyapunov residual $\|A_{\text{cl},j}^\top P_j + P_j A_{\text{cl},j} + Q_V\|_F = 3.77 \times 10^{-15}$. The closed-loop eigenvalues have real part -0.1620 , and the eigenvalues of P_j lie between 1.6128 and 35.6514. These checks verify the quadratic-CLF construction and the reduced-model input bound. They do not independently establish that every point in the full ellipsoid is collision-free; environmental safety remains a separate hard Poisson-HOCBF condition.

The experiment uses deterministic seed 7 and stores the effective configuration, scenario events, all state/control histories, certificate values, solver residuals, and a SHA-256 configuration hash. The reported QP filter requires approximately 0.25 ms at the 95th percentile, whereas complete guidance-field construction is approximately 1.18 s in the current Python implementation. Accordingly, the present evidence supports fast pointwise filtering with a precomputed static field; it does

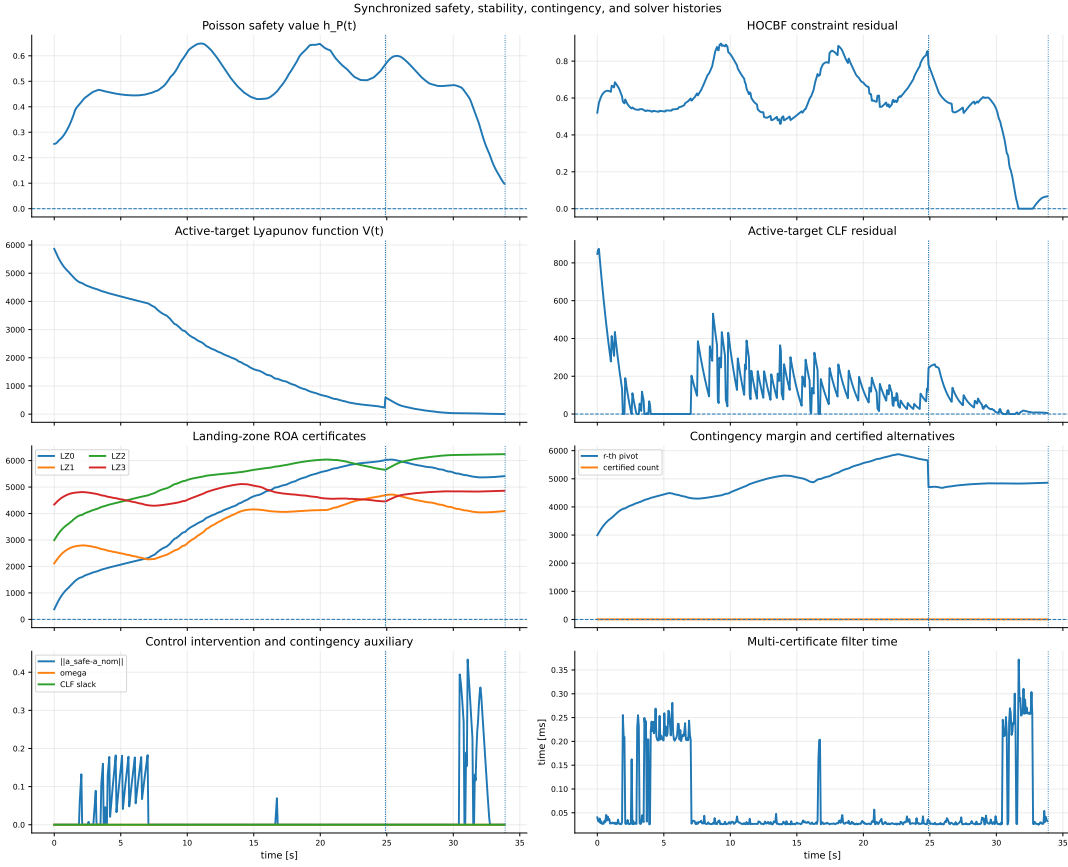


Fig. 11: Synchronized pointwise histories for the one-failure landing. The panels are computed from the local Poisson sample, HOCBF row, active-target CLF, four candidate certificates, shared contingency variable, control correction, and measured QP time at every integration step. The two vertical lines mark the target rejection/switch and the terminal landing event.

not yet support high-rate online reconstruction of the full three-dimensional Poisson field.

APPENDIX A AUXILIARY CERTIFICATE RESULTS

A. CLF Stabilization and Sublevel Invariance

Let V satisfy Definition 1 and let $\dot{V}(\mathbf{x}, \mathbf{k}(\mathbf{x})) \leq -\alpha_V(V(\mathbf{x}))$. The scalar comparison system

$$\dot{z} = -\alpha_V(z), \quad z(0) = V(\mathbf{x}(0)), \quad (94)$$

has a nonincreasing solution satisfying $z(t) \rightarrow 0$. The comparison lemma gives

$$0 \leq V(\mathbf{x}(t)) \leq z(t). \quad (95)$$

so $V(\mathbf{x}(t)) \rightarrow 0$. The bounds in (3a) then imply $\mathbf{x}(t) \rightarrow \mathbf{x}^*$.

For the sublevel set $\Omega_c = \{\mathbf{x} \mid V(\mathbf{x}) \leq c\}$, every boundary point satisfies

$$\dot{V}(\mathbf{x}, \mathbf{k}(\mathbf{x})) \leq -\alpha_V(c) < 0, \quad V(\mathbf{x}) = c. \quad (96)$$

Thus the closed-loop vector field points toward smaller values of V on $\partial\Omega_c$, and Ω_c is forward invariant. Since the CLF comparison argument applies to every initial condition in Ω_c , the set is contained in the region of attraction.

B. Quadratic CLF from the Lyapunov Equation

For the closed-loop linearization

$$\delta\dot{\mathbf{x}} = A_{cl}\delta\mathbf{x}, \quad (97)$$

let $P = P^\top \succ 0$ satisfy

$$A_{cl}^\top P + PA_{cl} = -Q, \quad Q = Q^\top \succ 0. \quad (98)$$

For $V(\delta\mathbf{x}) = \delta\mathbf{x}^\top P \delta\mathbf{x}$,

$$\dot{V} = \delta\dot{\mathbf{x}}^\top P \delta\mathbf{x} + \delta\mathbf{x}^\top P \delta\dot{\mathbf{x}} \quad (99)$$

$$= \delta\mathbf{x}^\top (A_{cl}^\top P + PA_{cl}) \delta\mathbf{x} \quad (100)$$

$$= -\delta\mathbf{x}^\top Q \delta\mathbf{x}. \quad (101)$$

The eigenvalue bounds

$$\lambda_{\min}(P)\|\delta\mathbf{x}\|^2 \leq V(\delta\mathbf{x}) \leq \lambda_{\max}(P)\|\delta\mathbf{x}\|^2 \quad (102)$$

and

$$\dot{V} \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V \quad (103)$$

establish exponential stability of the linear closed loop. If the nonlinear closed-loop dynamics are

$$\delta\dot{\mathbf{x}} = A_{cl}\delta\mathbf{x} + \mathbf{r}(\delta\mathbf{x}), \quad \frac{\|\mathbf{r}(\delta\mathbf{x})\|}{\|\delta\mathbf{x}\|} \rightarrow 0, \quad (104)$$

then

$$\dot{V} = -\delta \mathbf{x}^\top Q \delta \mathbf{x} + 2\delta \mathbf{x}^\top P \mathbf{r}(\delta \mathbf{x}). \quad (105)$$

The second term is higher order and is dominated by the negative quadratic term on a sufficiently small neighborhood.

C. CBF Invariance

Assume h is a CBF and

$$\dot{h}(\mathbf{x}, \mathbf{k}(\mathbf{x})) \geq -\gamma(h(\mathbf{x})). \quad (106)$$

At a boundary point $h(\mathbf{x}) = 0$, the extended class- $\mathcal{K}$ property gives $\gamma(0) = 0$, and therefore

$$\dot{h}(\mathbf{x}, \mathbf{k}(\mathbf{x})) \geq 0. \quad (107)$$

The closed-loop vector field lies in the tangent cone of the zero-superlevel set. Nagumo's theorem then establishes forward invariance under the regularity assumptions stated in Definition 2 [4].

D. Relative-Degree-Two HOCBF Derivation

For

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{a}, \quad (108)$$

and $h_P = h_P(\mathbf{p})$, the chain rule gives

$$\dot{h}_P = Dh_P(\mathbf{p}) \mathbf{v}. \quad (109)$$

Differentiating again,

$$\ddot{h}_P = \frac{d}{dt}(Dh_P(\mathbf{p}) \mathbf{v}) \quad (110)$$

$$= \mathbf{v}^\top D^2 h_P(\mathbf{p}) \mathbf{v} + Dh_P(\mathbf{p}) \mathbf{a}. \quad (111)$$

Define

$$h_0 = h_P, \quad h_1 = \dot{h}_P + \gamma_1 h_P. \quad (112)$$

The HOCBF condition $\dot{h}_1 + \gamma_2 h_1 \geq 0$ expands as

$$Dh_P(\mathbf{p}) \mathbf{a} + \mathbf{v}^\top D^2 h_P(\mathbf{p}) \mathbf{v} + (\gamma_1 + \gamma_2) Dh_P(\mathbf{p}) \mathbf{v} + \gamma_1 \gamma_2 h_P(\mathbf{p}) \geq 0, \quad (113)$$

which yields (28).

E. Projection onto a Single CBF Half-Space

Consider

$$\begin{aligned} \min_{\mathbf{u}} \quad & \frac{1}{2} \|\mathbf{u} - \mathbf{u}_{\text{nom}}\|_2^2 \\ \text{s.t.} \quad & \mathbf{a}^\top \mathbf{u} \geq b, \quad \mathbf{a} \neq 0. \end{aligned} \quad (114)$$

The Lagrangian is

$$\mathcal{L}(\mathbf{u}, \lambda) = \frac{1}{2} \|\mathbf{u} - \mathbf{u}_{\text{nom}}\|_2^2 + \lambda(b - \mathbf{a}^\top \mathbf{u}), \quad \lambda \geq 0. \quad (115)$$

Stationarity gives

$$\mathbf{u} = \mathbf{u}_{\text{nom}} + \lambda \mathbf{a}. \quad (116)$$

If the nominal input is feasible, complementary slackness gives $\lambda = 0$. Otherwise the constraint is active and

$$\lambda = \frac{b - \mathbf{a}^\top \mathbf{u}_{\text{nom}}}{\|\mathbf{a}\|^2}. \quad (117)$$

Combining both cases yields

$$\mathbf{u}_{\text{safe}} = \mathbf{u}_{\text{nom}} + \frac{\max\{0, b - \mathbf{a}^\top \mathbf{u}_{\text{nom}}\}}{\|\mathbf{a}\|^2} \mathbf{a}. \quad (118)$$

F. Combinatorial Stabilization Condition

Let $\{V_j\}_{j \in [p]}$ be local CLFs about equilibria $\{\mathbf{x}_j^*\}_{j \in [p]}$, let $h_j(\mathbf{x}) = c_j - V_j(\mathbf{x})$, and let $\tilde{h}_r(\mathbf{x}) = \max^{(r)}\{h_j(\mathbf{x})\}_{j \in [p]}$. Fix a selected equilibrium index $j^\dagger \in [p]$. Suppose there exist a continuous feedback $\mathbf{k}$ and a nonnegative auxiliary function θ such that

$$\begin{aligned} \dot{V}_{j^\dagger}(\mathbf{x}, \mathbf{k}(\mathbf{x})) &\leq -\alpha_{j^\dagger}(V_{j^\dagger}(\mathbf{x})) \\ &\quad + \theta(\mathbf{x}) \text{ReLU}(-h_{j^\dagger}(\mathbf{x})), \end{aligned} \quad (119a)$$

$$\begin{aligned} \dot{h}_j(\mathbf{x}, \mathbf{k}(\mathbf{x})) &\geq -\alpha_c(h_j(\mathbf{x})) \\ &\quad - \theta(\mathbf{x}) \rho(h_j(\mathbf{x}) - \tilde{h}_r(\mathbf{x})), \end{aligned} \quad (119b)$$

for every $j \in [p]$. The second inequality renders the combinatorial region $\tilde{\mathcal{R}}_r$ forward invariant by the combinatorial CBF result. Inside the selected certified region, $h_{j^\dagger} \geq 0$ and therefore $\text{ReLU}(-h_{j^\dagger}) = 0$; the first inequality reduces to the ordinary CLF decrease condition. Consequently, trajectories initialized in the selected certified component converge to $\mathbf{x}_{j^\dagger}^*$ while remaining in the region of attraction of at least r candidates. This is the central implication used by the combinatorial stabilization theorem in [9].

G. Combinatorial Set Equivalence

Let $h_{(1)}(\mathbf{x}) \geq \dots \geq h_{(p)}(\mathbf{x})$ be the sorted certificate values. By definition,

$$\tilde{h}_r(\mathbf{x}) = h_{(r)}(\mathbf{x}). \quad (120)$$

If $h_{(r)}(\mathbf{x}) \geq 0$, then the first r sorted values are nonnegative, so at least r certificates are valid. Conversely, if at least r certificates are nonnegative, then the r -th largest value is nonnegative. Therefore,

$$\tilde{h}_r(\mathbf{x}) \geq 0 \iff |\{j \in [p] \mid h_j(\mathbf{x}) \geq 0\}| \geq r. \quad (121)$$

The pivot is continuous but generally nondifferentiable where two certificates exchange order. The smooth combinatorial rows in (35) avoid differentiating the pivot itself [10].

APPENDIX B

ELLIPTIC PRELIMINARIES FOR POISSON SAFETY FUNCTIONS

A. Hölder Spaces

For $\alpha \in (0, 1)$, a function $f : \bar{\Omega} \rightarrow \mathbb{R}$ is Hölder continuous with exponent α if

$$[f]_{C^{0,\alpha}(\bar{\Omega})} := \sup_{\substack{\mathbf{y}_1, \mathbf{y}_2 \in \bar{\Omega} \\ \mathbf{y}_1 \neq \mathbf{y}_2}} \frac{|f(\mathbf{y}_1) - f(\mathbf{y}_2)|}{\|\mathbf{y}_1 - \mathbf{y}_2\|^\alpha} < \infty. \quad (122)$$

The space $C^{k,\alpha}(\bar{\Omega})$ contains functions whose partial derivatives through order k are continuous and whose derivatives of order k are Hölder continuous.

B. Minimum Principles

Let $h \in C^2(\Omega) \cap C^0(\bar{\Omega})$. If

$$\Delta h \leq 0 \quad \text{in } \Omega, \quad (123)$$

then h is superharmonic and the weak minimum principle gives

$$\min_{\bar{\Omega}} h = \min_{\partial\Omega} h. \quad (124)$$

For the Poisson problem with $f_P < 0$ and zero Dirichlet data,

$$h_P \geq 0 \quad \text{on } \bar{\Omega}. \quad (125)$$

The strong minimum principle excludes an interior zero for a nonconstant strictly superharmonic solution. Hence,

$$h_P > 0 \quad \text{in } \Omega. \quad (126)$$

C. Hopf Boundary-Point Lemma

Let $\mathbf{y}_0 \in \partial\Omega$ satisfy the interior-sphere condition and let $\hat{\mathbf{n}}(\mathbf{y}_0)$ be the outward unit normal of the free-space domain. If a nonconstant superharmonic function attains its minimum at $\mathbf{y}_0$, Hopf's lemma gives

$$Dh_P(\mathbf{y}_0) \cdot \hat{\mathbf{n}}(\mathbf{y}_0) < 0. \quad (127)$$

Consequently,

$$Dh_P(\mathbf{y}_0) \neq 0. \quad (128)$$

This supplies the regular zero-level-set condition used by the CBF construction. At nonsmooth corners, the classical pointwise normal result requires additional geometric or weak-solution arguments [15].

D. Elliptic Regularity

Under the standard global Schauder assumptions on the boundary and Dirichlet data,

$$f_P \in C^{k,\alpha}(\bar{\Omega}) \implies h_P \in C^{k+2,\alpha}(\bar{\Omega}). \quad (129)$$

Thus a sufficiently regular forcing function and boundary produce the first and second derivatives required by velocity-level CBFs and acceleration-level HOCBFs. Interior regularity requires weaker boundary assumptions, but controller evaluation near obstacle surfaces requires regularity up to the boundary [15], [16].

E. Proof of the Poisson Safety-Function Theorem

Existence and uniqueness follow from the classical Dirichlet theory under the assumptions of Theorem 1. The Schauder estimate gives

$$h_P \in C^{k+2,\alpha}(\bar{\Omega}). \quad (130)$$

Since $f_P < 0$, the solution is strictly superharmonic. The weak and strong minimum principles yield

$$h_P \geq 0 \quad \text{on } \bar{\Omega}, \quad h_P > 0 \quad \text{in } \Omega. \quad (131)$$

Hopf's lemma yields a nonzero normal derivative at regular boundary points. Therefore the zero-superlevel set equals $\bar{\Omega}$ and the zero level set is regular, proving that h_P is a safety function of order $k+2$ [5].

F. Dirichlet Principle

Writing $q = -f_P > 0$, the equation is

$$-\Delta h_P = q, \quad h_P|_{\partial\Omega} = 0. \quad (132)$$

The weak form seeks $h_P \in H_0^1(\Omega)$ such that

$$\int_{\Omega} Dh_P(\mathbf{y}) D\varphi(\mathbf{y})^\top dV = \int_{\Omega} q(\mathbf{y}) \varphi(\mathbf{y}) dV \quad (133)$$

for every $\varphi \in H_0^1(\Omega)$. Equivalently, h_P minimizes

$$J[h] = \int_{\Omega} \left(\frac{1}{2} \|Dh(\mathbf{y})\|^2 + h(\mathbf{y}) f_P(\mathbf{y}) \right) dV \quad (134)$$

over admissible functions satisfying the homogeneous boundary condition. The Euler–Lagrange equation of this functional is the Poisson equation [16].

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